

CAT 2025

MBA FASTRACK



Lecture - 1

Geometry

2D Geometry - 1

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RECAP OF THE PREVIOUS LECTURE

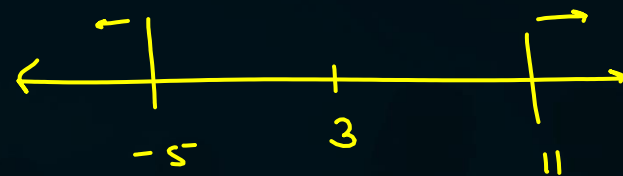


✓ $|8| = 8$ *as it is*

✓ $|-8| = 8$

$-(-8)$

$$|x - 3| \geq 8$$



$$[-\infty, -5] \cup [11, \infty)$$

$$|x - 2| + |x - 5| = 7$$



$$|2x - 3| = |3 - 2x|$$

TOPICS *to be covered*



Topic

Modulus – Graph

Topic

Geometry – Introduction

Topic

Triangles – Basic Properties

Topic

Right Angle Triangle + Equilateral Triangle

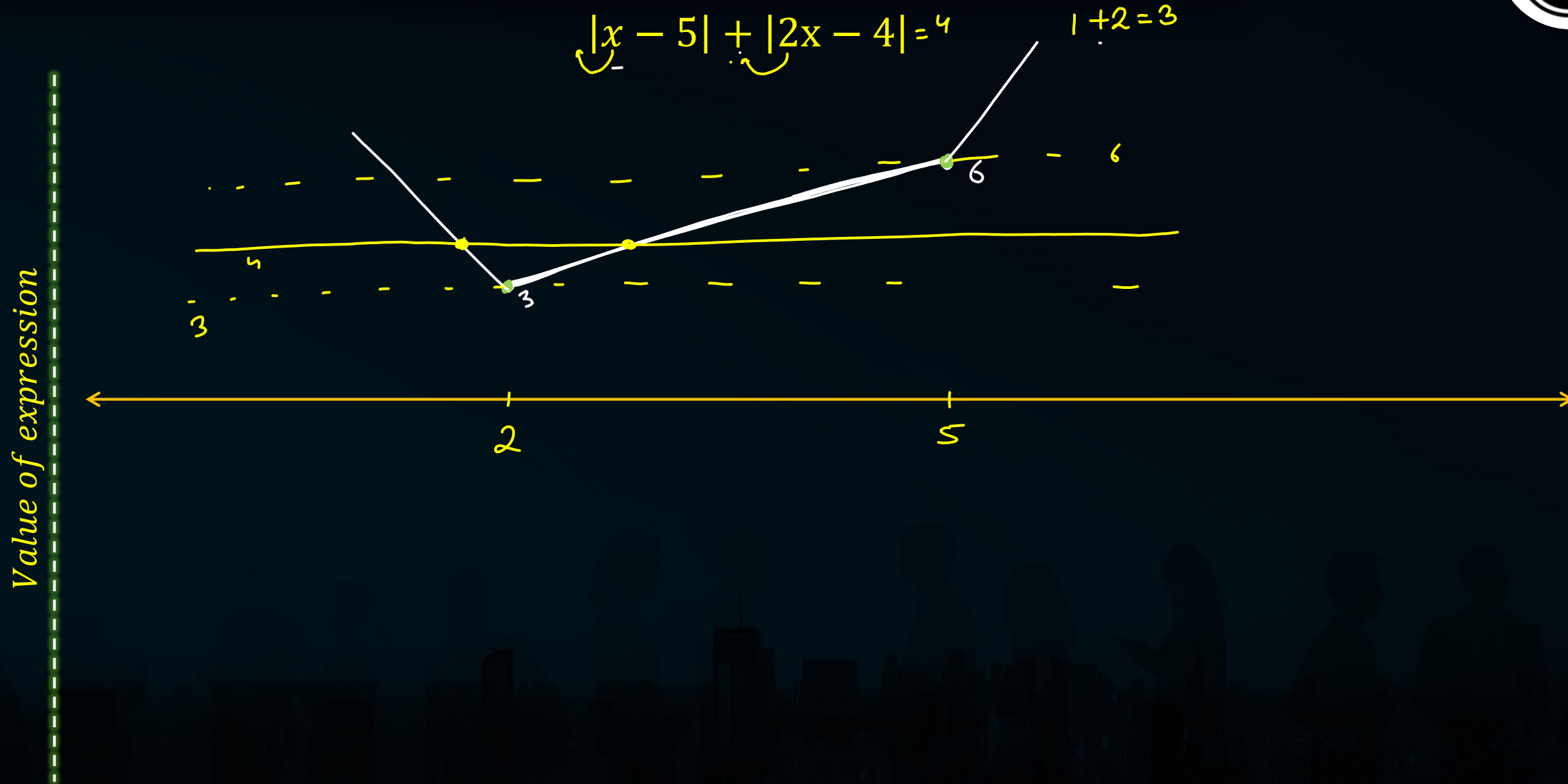
Topic

Important Lines and Centers



Topics : Modulus Graph

Concept



Concept



$$1 + 2 - 3 = 0$$

$$|x - 5| + |2x - 4| - 3|1 - x| = k$$

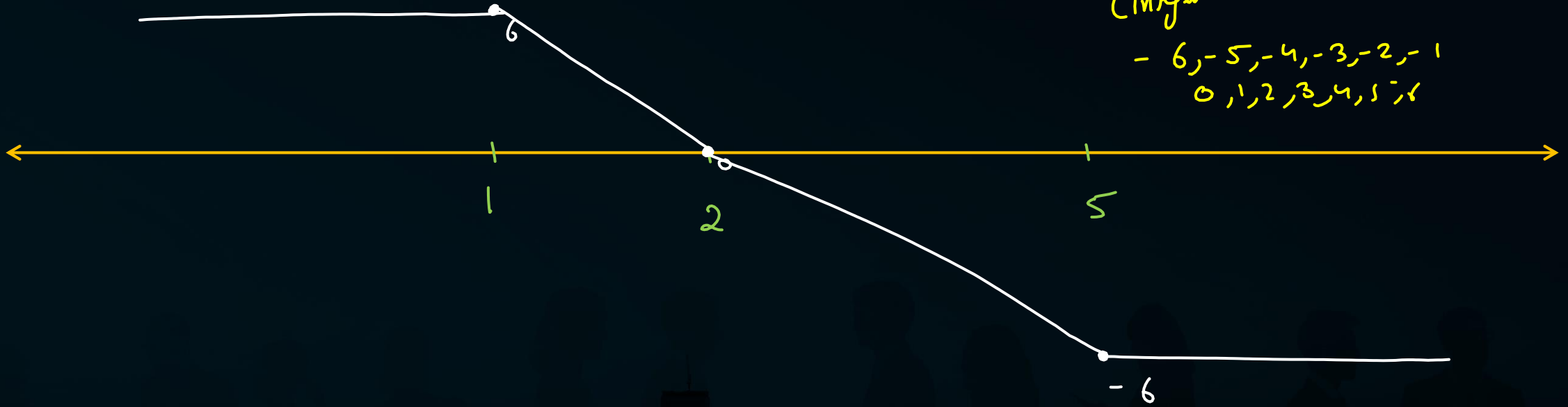
$$|x - 5| + |2x - 4| - 3|x - 1| \rightarrow \text{Range} = [-6, 6]$$

$$k = \underline{\underline{13}}$$

(integers)

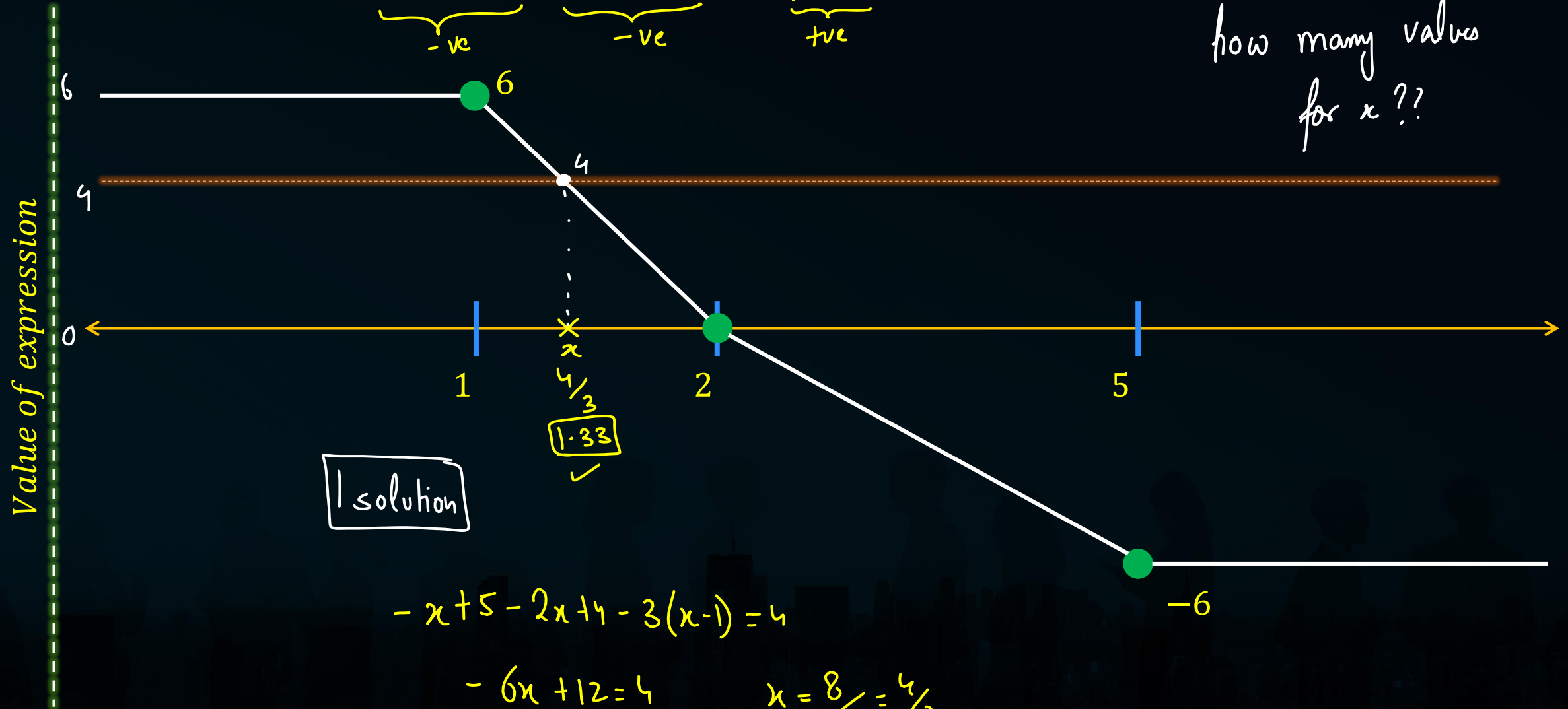
-6, -5, -4, -3, -2, -1
0, 1, 2, 3, 4, 5, 6

Value of expression



$$\underbrace{|x-5|}_{-ve} + \underbrace{|2x-4|}_{-ve} - 3\underbrace{|x-1|}_{+ve} = 4$$

how many values
for x ??



1 solution

$$-x + 5 - 2x + 4 - 3(x-1) = 4$$

$$-6x + 12 = 4$$

$$-6x = -8$$

$$x = \frac{8}{6} = \frac{4}{3}$$

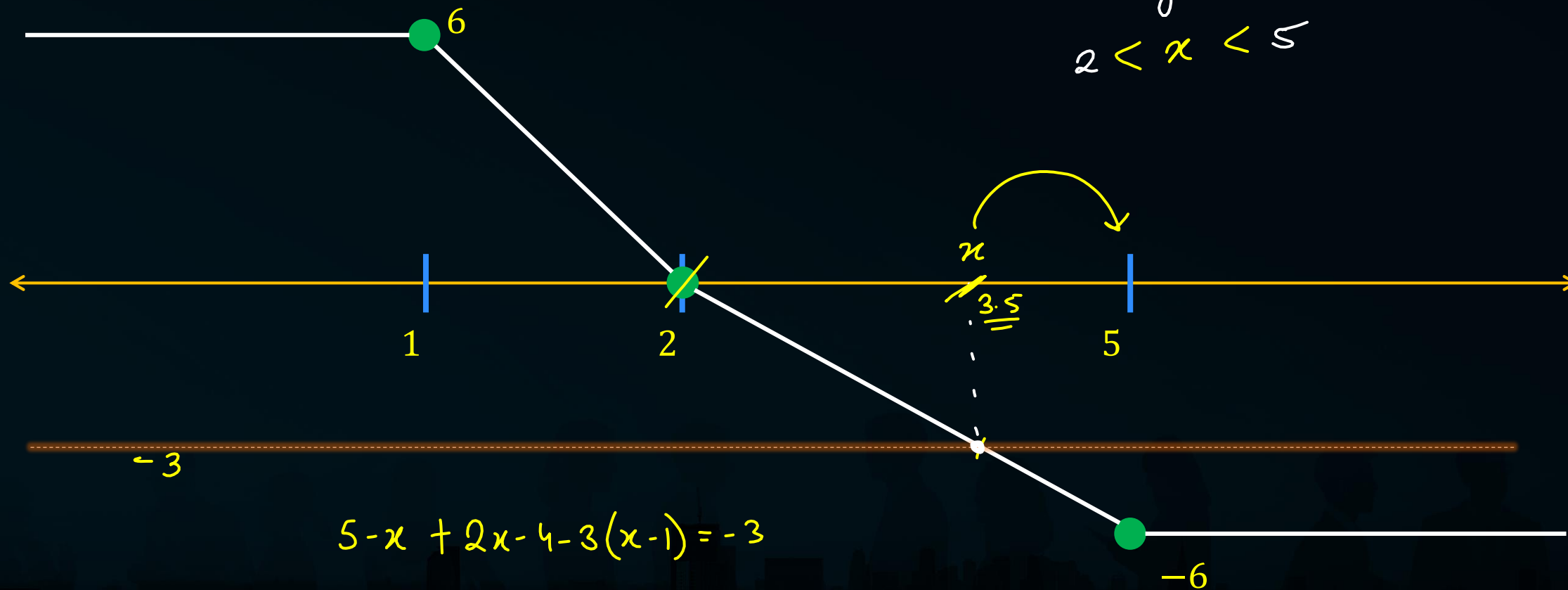
Questions 1



#Q: $\underbrace{|x - 5|} + |2x - 4| - 3|x - 1| = -3$

No. of solutions. = 1
 $2 < x < 5$

Value of expression



$$5 - x + 2x - 4 - 3(x - 1) = -3$$

$$-2x + 4 = -3$$

$$-2x = -7$$

$$\boxed{x = 3.5}$$

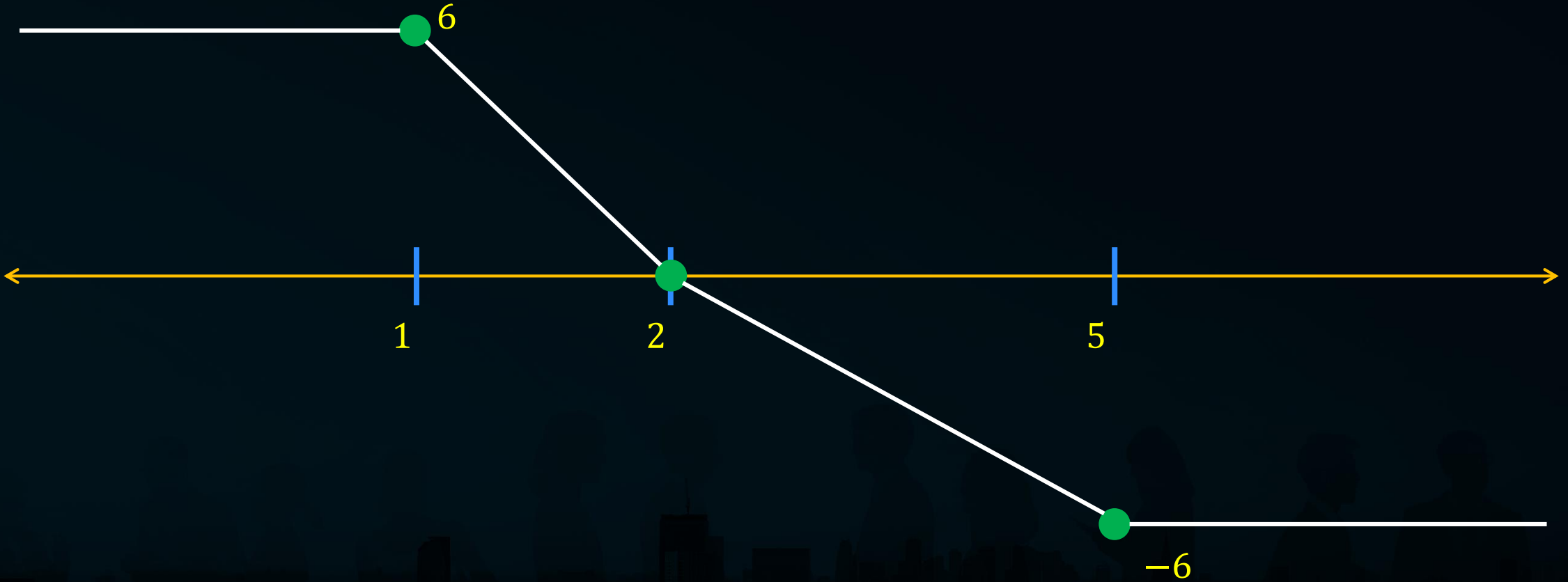
Questions 2



#Q: $|x - 5| + |2x - 4| - 3|x - 1| = 8$

No. of solutions = 0

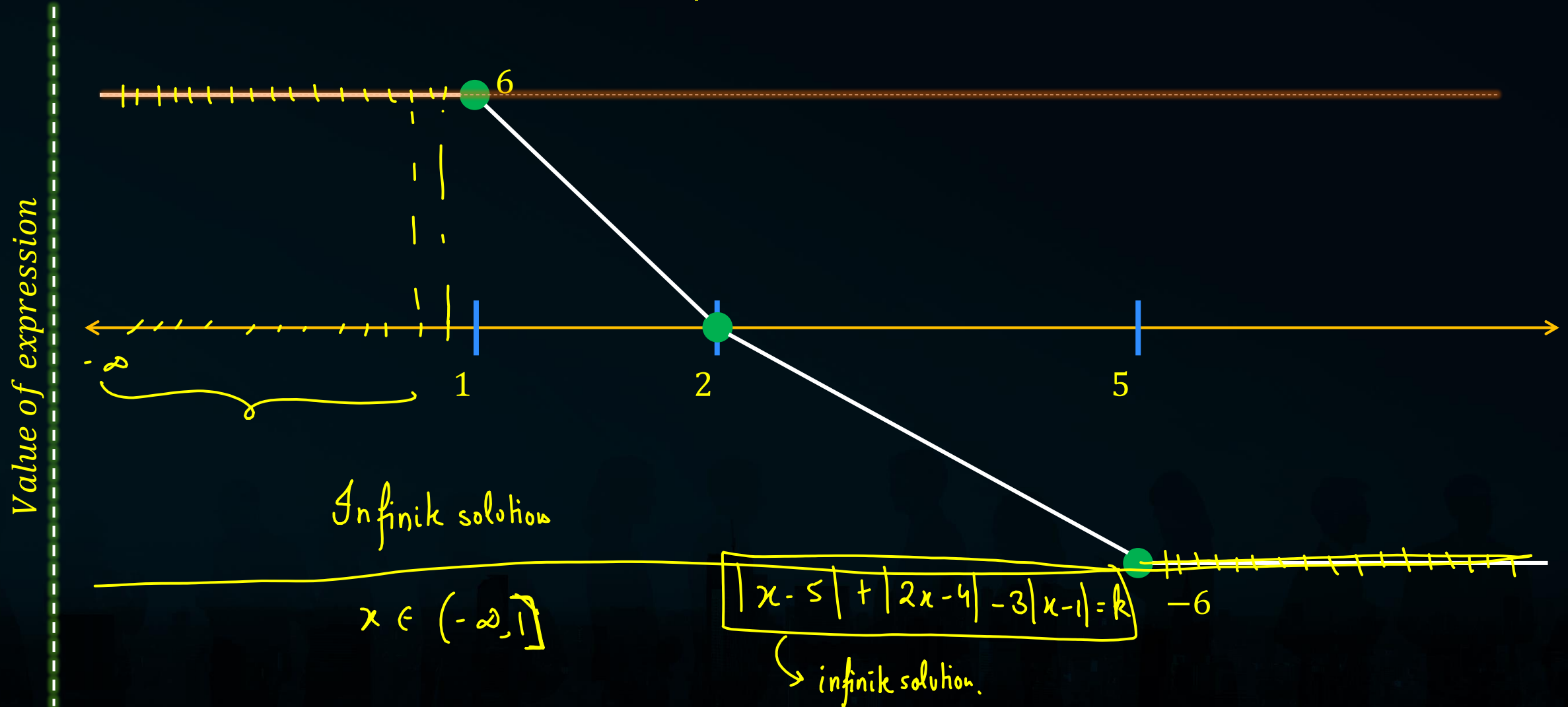
Value of expression



Questions 3



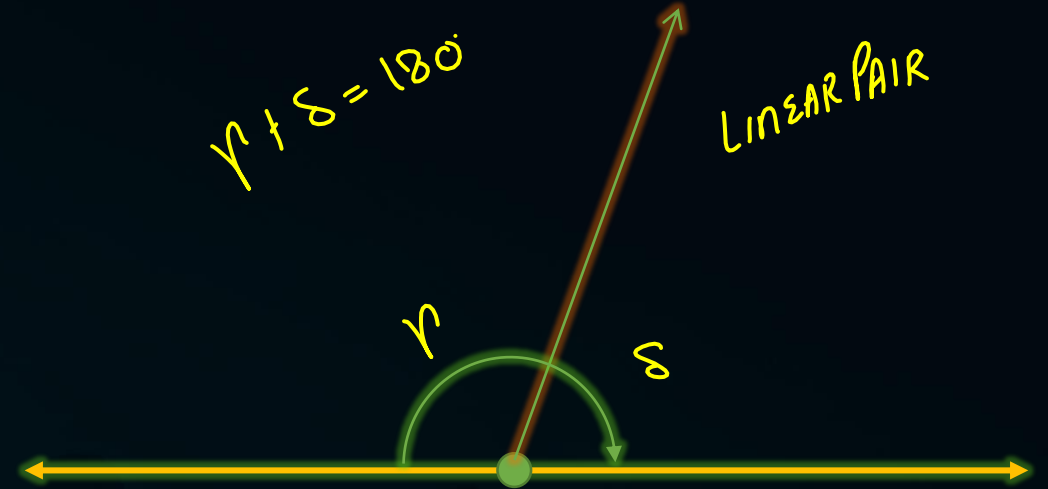
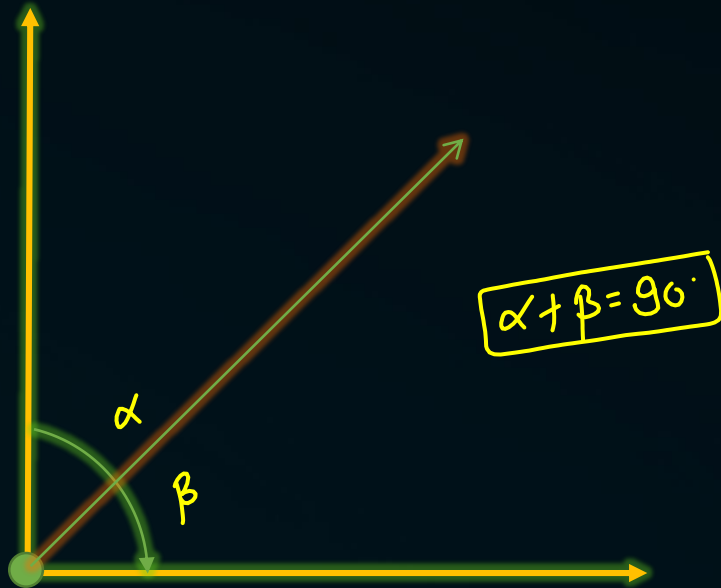
#Q: $|x - 5| + |2x - 4| - 3|x - 1| = 6$





Topics : Geometry Introduction

Complimentary and Supplementary Angles



- Two angles are said to be **complimentary** if the **sum of the angles is equal to 90°** .
- Two angles are said to be **supplimentary** if the **sum of the angles is equal to 180°** .

QUESTION-4



#Q. Find $y + x$.

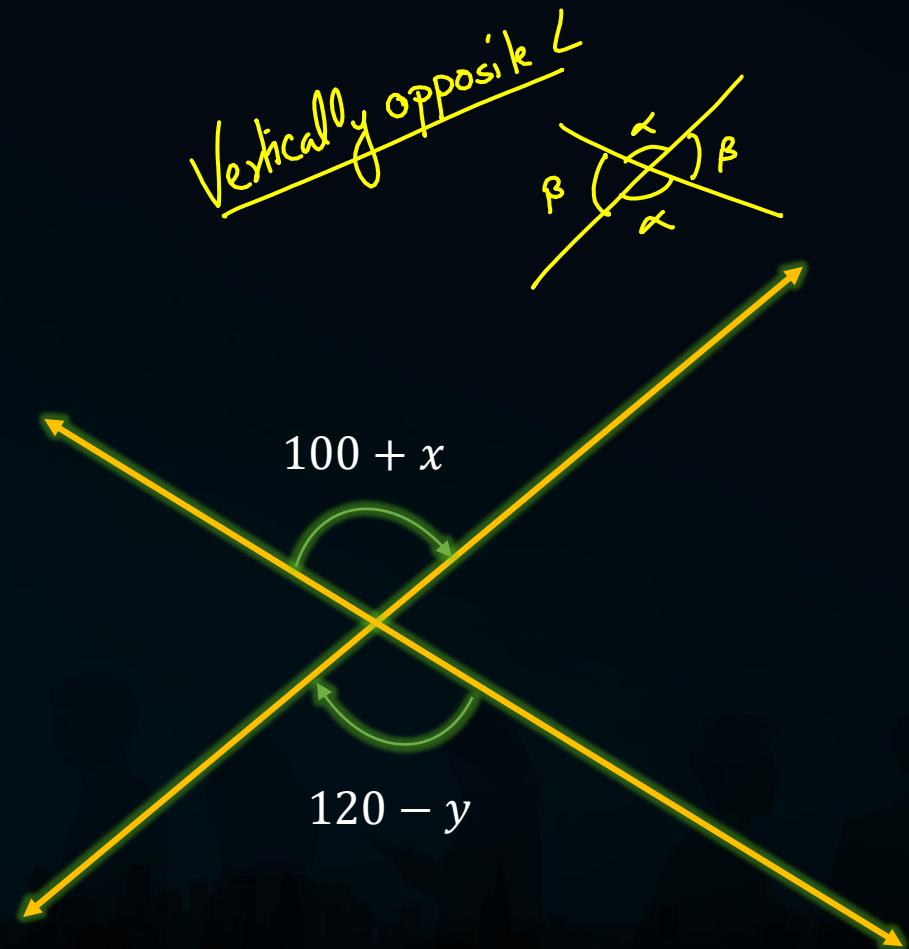
A 20 ✓✓

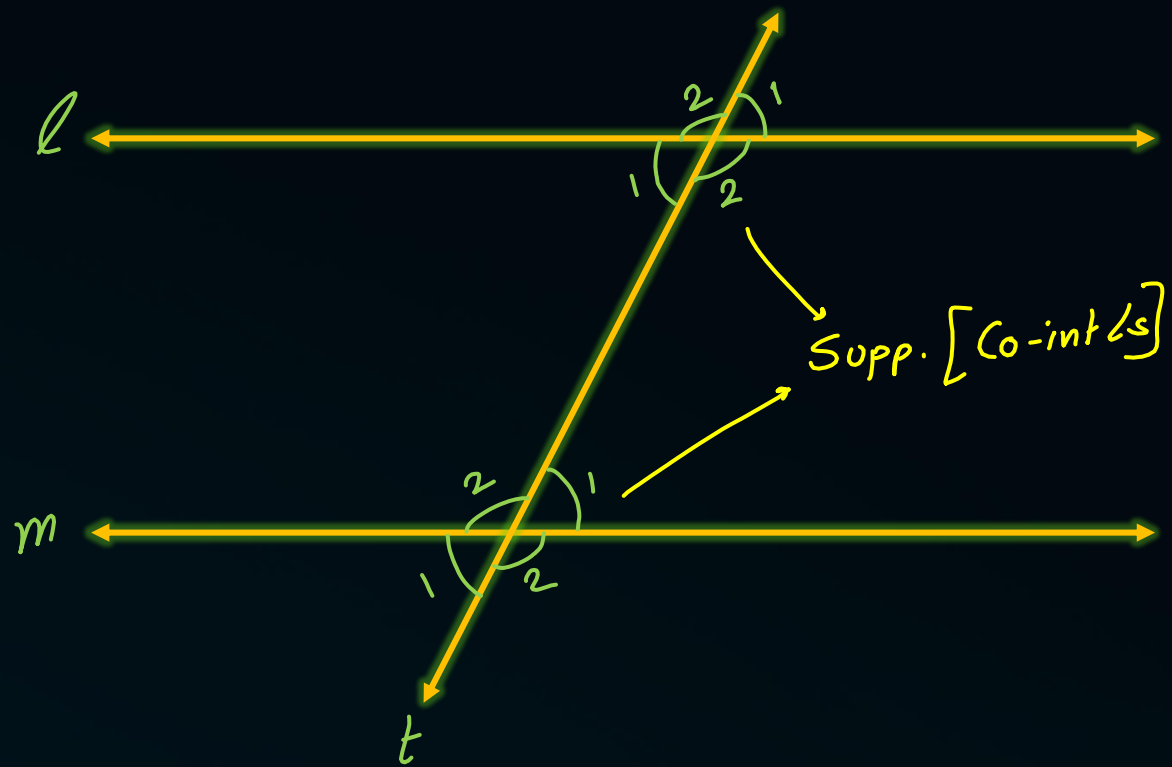
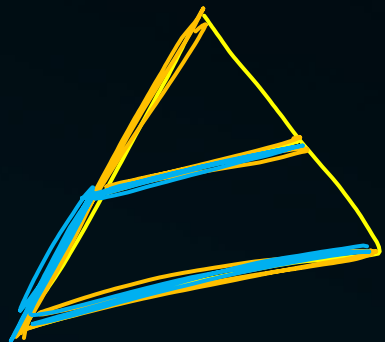
B 220

C 230

D 40

$$\begin{aligned} 100 + x &= 120 - y \\ x + y &= 120 - 100 \\ &= 20 \end{aligned}$$





Corresponding $\angle$ s



$$\angle 1 + \angle 2 = 180^\circ$$



Alt $\angle$ s

QUESTION-5



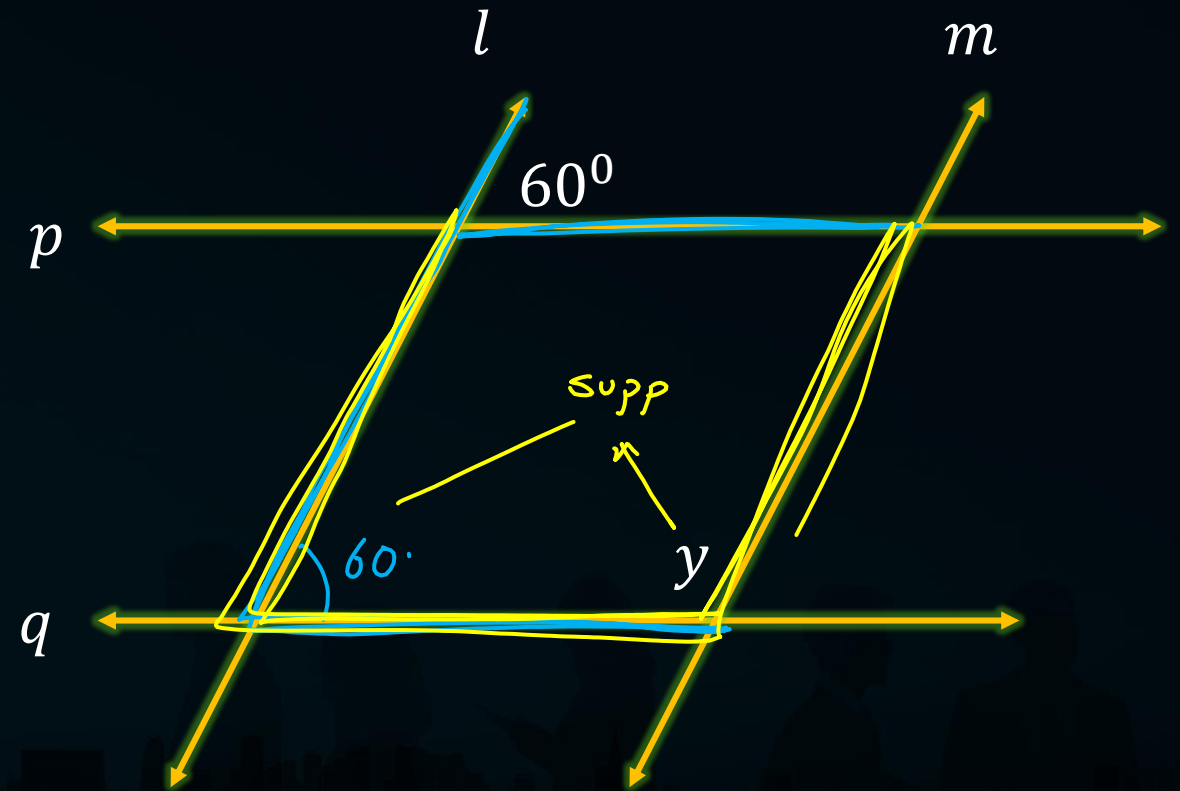
#Q. If 'l' is Parallel to 'm' and 'p' is parallel to 'q', find 'y'.

A 30°

B 60°

C 90°

D 120° ✓

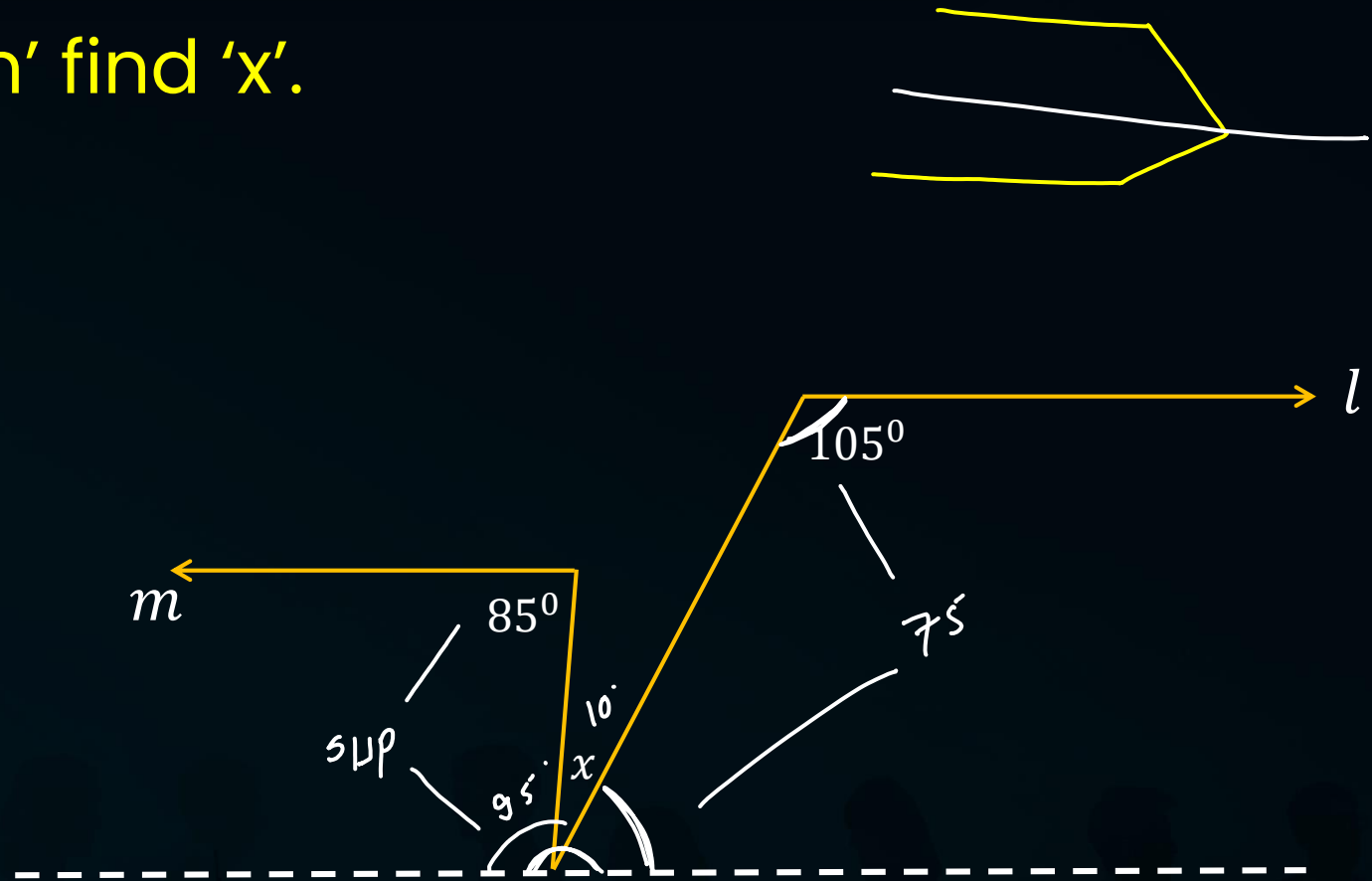


QUESTION-6



#Q. If 'l' is Parallel to 'm' find 'x'.

- A** 10° ✓
- B** 15°
- C** 20°
- D** 25°



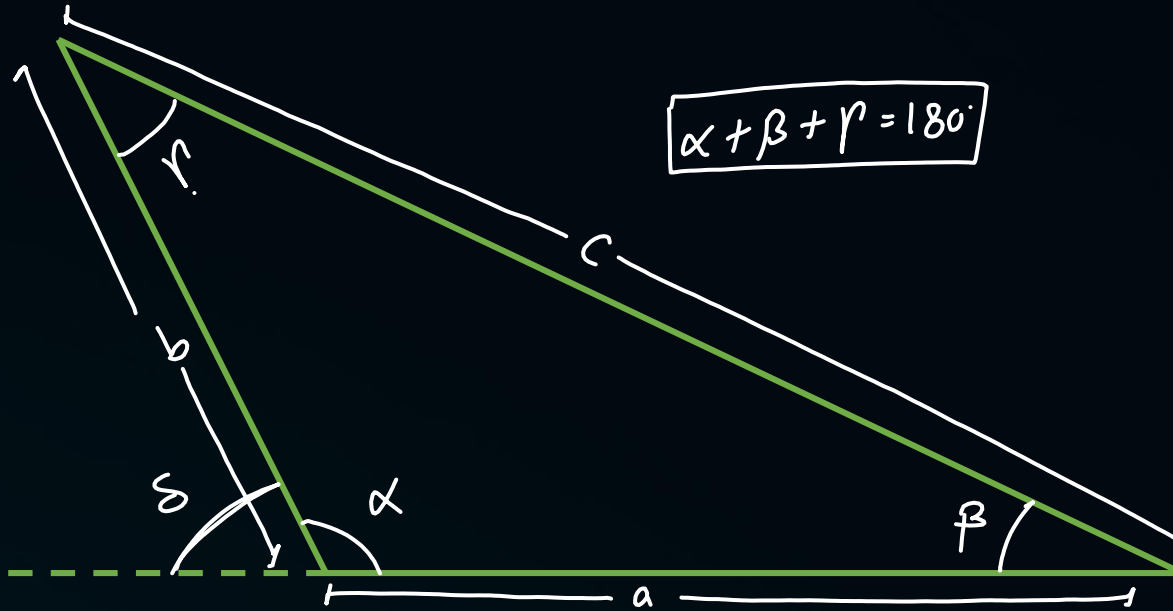


Topics : Triangle Basic Properties

Key Properties

$$s = \frac{a+b+c}{2}$$

$$\begin{array}{l} a < s \\ b < s \\ c < s \end{array}$$



❖ SUM OF ANGLES OF A TRIANGLE = 180°

❖ EXTERIOR ANGLE PROPERTY: exterior angle = sum of interior opposite angles

❖ Sum of two sides > Third side

$$a + b > c$$

$$b + c > a$$

$$c + a > b$$

❖ Third Side > Difference of Two Sides

$$c > |a - b|$$

$$b > |a - c|$$

$$a > |b - c|$$

❖ Any side < Semi perimeter

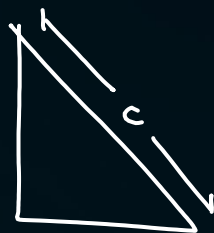
$$\left. \begin{array}{l} 2, 3, 5 \\ 2 + 3 \neq 5 \end{array} \right\} \times$$

[if c is longest side]

$$a^2 + b^2 = c^2 \longrightarrow \text{Right } \angle \Delta$$

$$a^2 + b^2 < c^2 \longrightarrow \text{obtuse } \angle \Delta$$

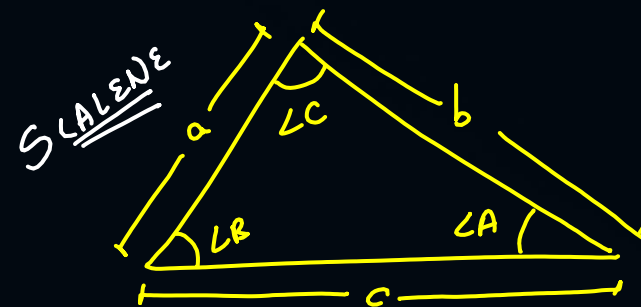
$$a^2 + b^2 > c^2 \longrightarrow \text{Acute } \angle \Delta$$



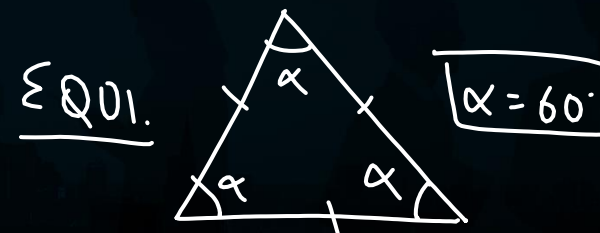
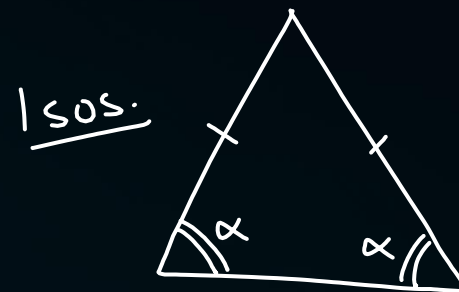
$$3, 5, 7$$

$$\underbrace{3^2 + 5^2}_{34} < \underbrace{7^2}_{49}$$

$\longrightarrow$ obtuse



$$\boxed{\begin{array}{l} c > b > a \\ \angle C > \angle B > \angle A \end{array}}$$

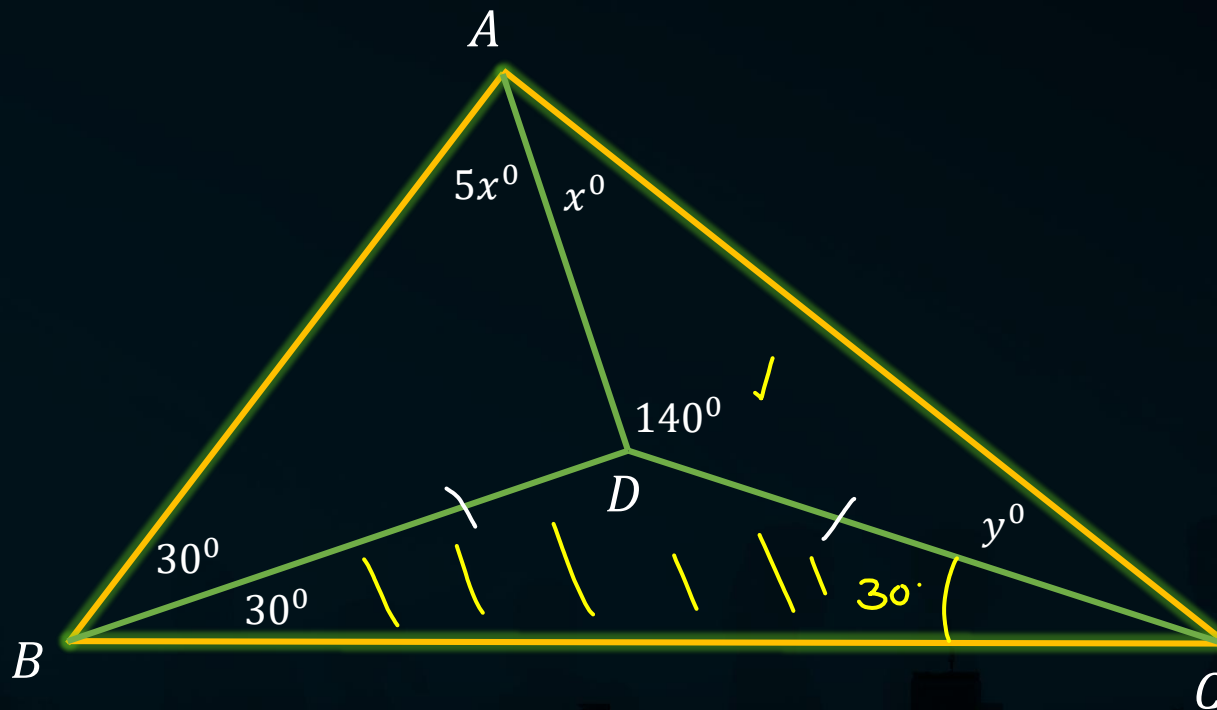


$$\boxed{\alpha = 60^\circ}$$

QUESTION-7



#Q. In the figure below, $DB = DC$. What is the value of 'y'?



$$140 + x + y = 180$$

$$x + y = 40$$

$$y = 30$$

$$6x + 60 + (30 + y) = 180$$

$$6x + y = 90$$

$$5x = 50$$

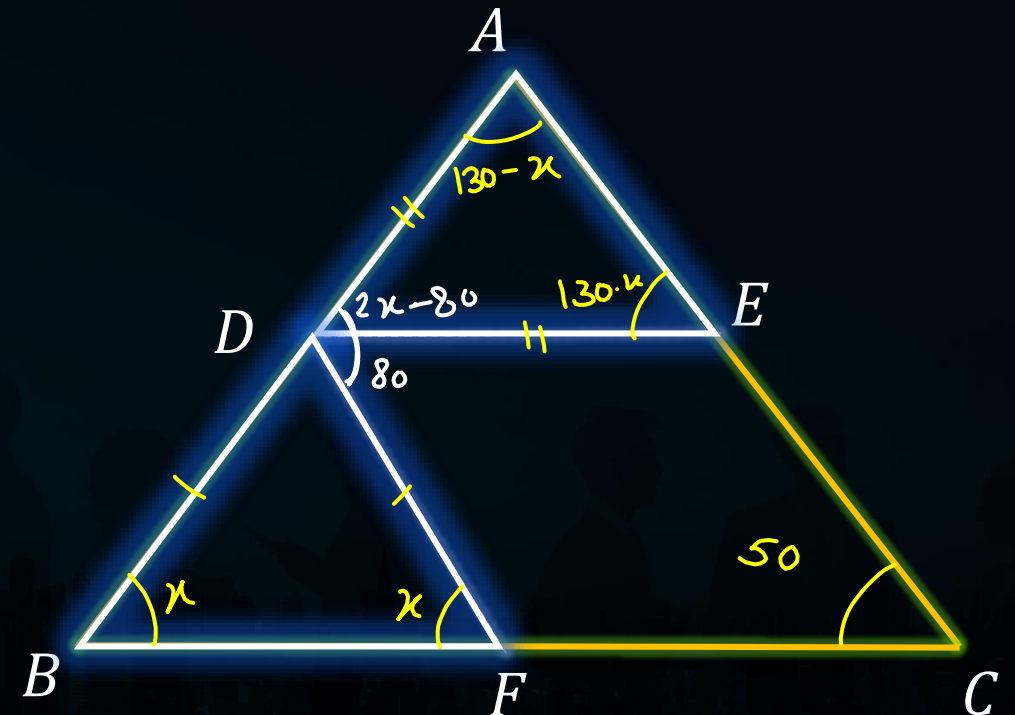
$$x = 10$$

QUESTION-8 (CAT 2021 Slot 3)



#Q. In a triangle ABC , $\angle BCA = 50^\circ$. D and E are points on AB and AC , respectively, such that $AD = DE$. If F is a point on BC such that $BD = DF$, then $\angle FDE$, in degrees, is equal to

- (A) 100°
- (B) 96°
- (C) 80° ✓
- (D) 72°

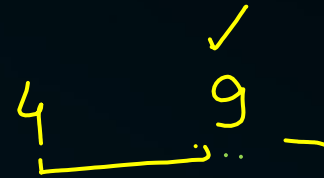


QUESTION - 9



#Q. If two sides of a triangle are 4 and 9, find the number of possible integral values 3rd side can take?

$$a^2 + b^2 = c^2$$



$$5 < c < 13$$

6, 7, 8, 9, 10, 11, 12

$c^2 = 81$

ob	4, 6, 9
ob	4, 7, 9
ob	4, 8, 9
ac	4, 9, 9

4, 9, 10
4, 9, 11
4, 9, 12

ob

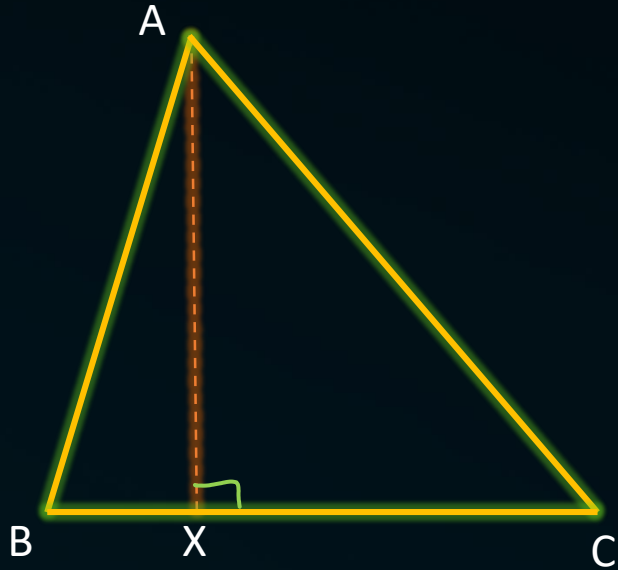
6 obtuse 1 acute

7 possible Δ s

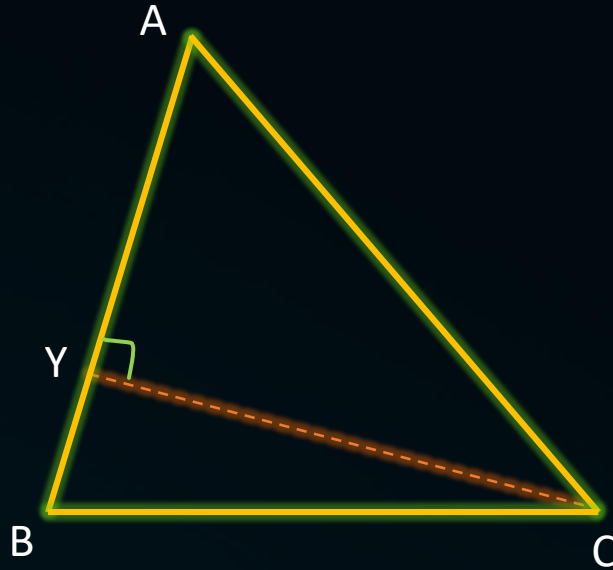


Area of Triangle

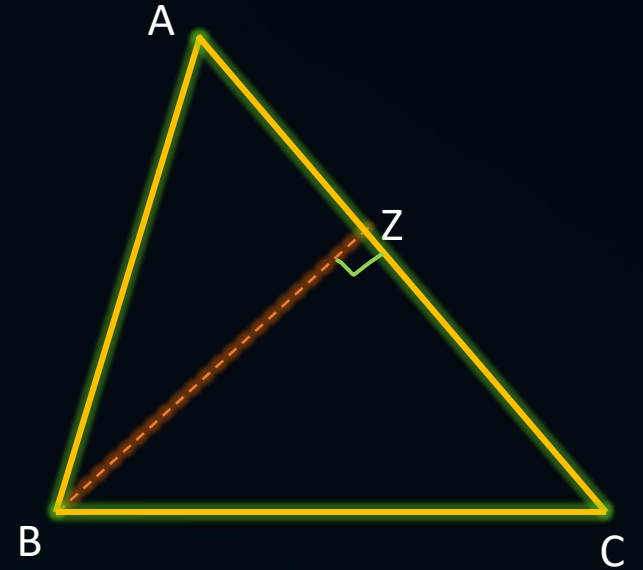
Area of Any Triangle



$$\frac{1}{2} (BC)(AX)$$



$$\frac{1}{2} (AB)(CY)$$

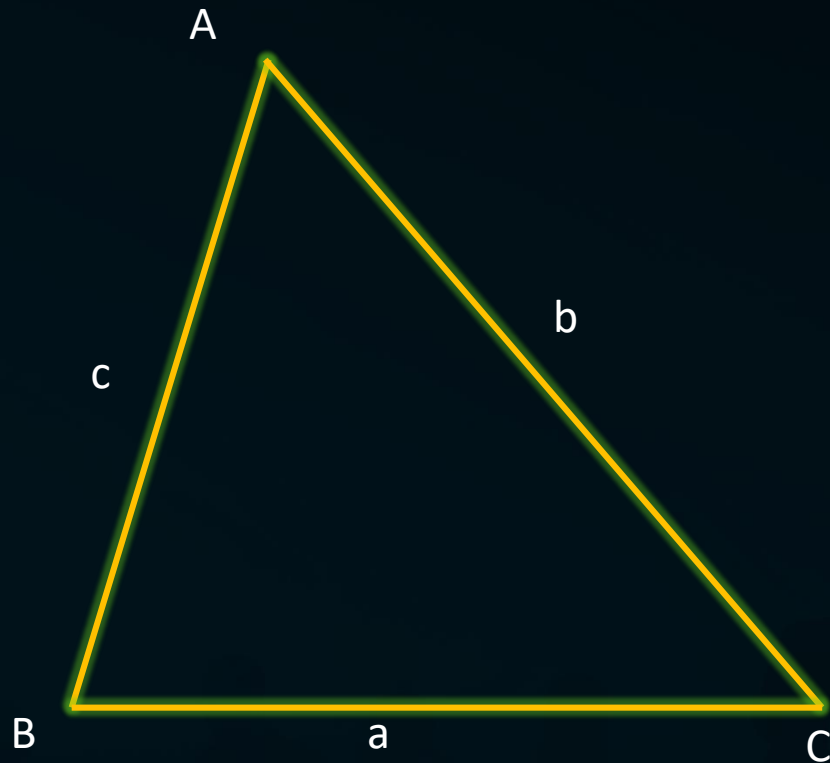


$$\frac{1}{2} (AC)(BZ)$$

$$\text{Area of Triangle } ABC = \frac{1}{2} (BC)(AX) = \frac{1}{2} (AB)(CY) = \frac{1}{2} (AC)(BZ)$$

$$(BC)(AX) = (AB)(CY) = (AC)(BZ)$$

Heron's Formula



$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$s = \frac{a+b+c}{2}$$

True for All Δ s

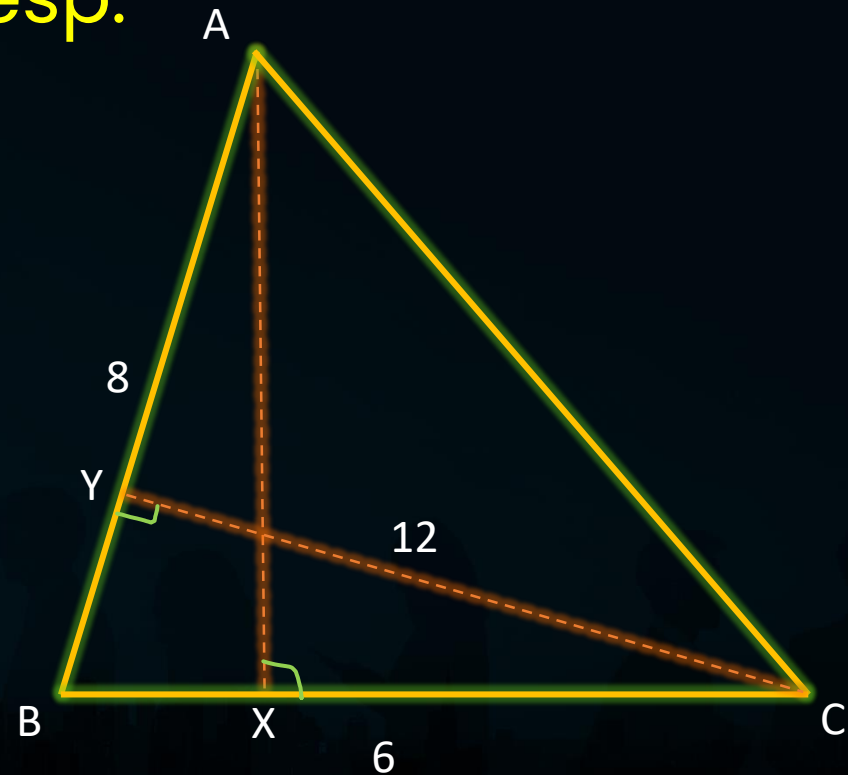
QUESTION - 10



#Q. In the following figure, $AB = 8\text{cm}$, $BC = 6\text{cm}$, $CY = 12\text{cm}$. find the length of AX given that AX and CY are perpendicular to BC and AB resp.

- A** 8 cm
- B** 14 cm
- C** 16 cm ✓✓
- D** 10 cm

$$6(AX) = 8(12)$$
$$\underline{AX = 16}$$



QUESTION - 11



#Q. In the following figure, $AB = 8\text{cm}$, $BC = 5\text{cm}$, $CA = 11\text{cm}$. find the length of AX .

A $\frac{8}{5}\sqrt{14}$

$$\sqrt{\frac{12(4)(1)(7)}{4(3)}} = \frac{1}{2}(5)(AX)$$

B $\frac{8}{5}\sqrt{21}$ ✓✓

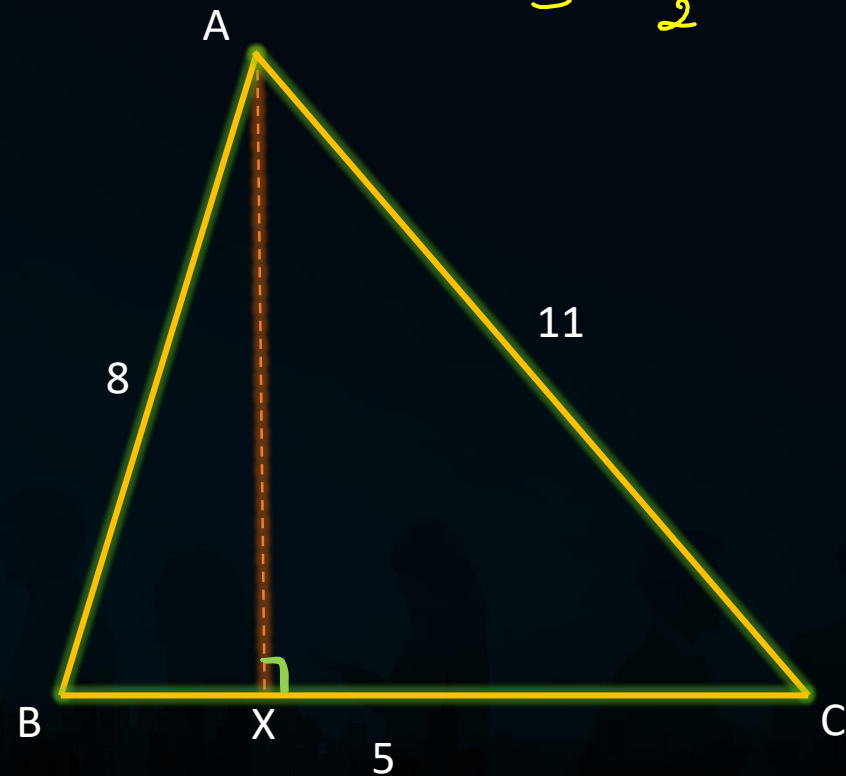
$$4\sqrt{21} = \frac{1}{2}(5)(AX)$$

$$\frac{8}{5}\sqrt{21} = AX$$

C $\frac{3}{5}\sqrt{7}$

D $\frac{8}{5}\sqrt{28}$

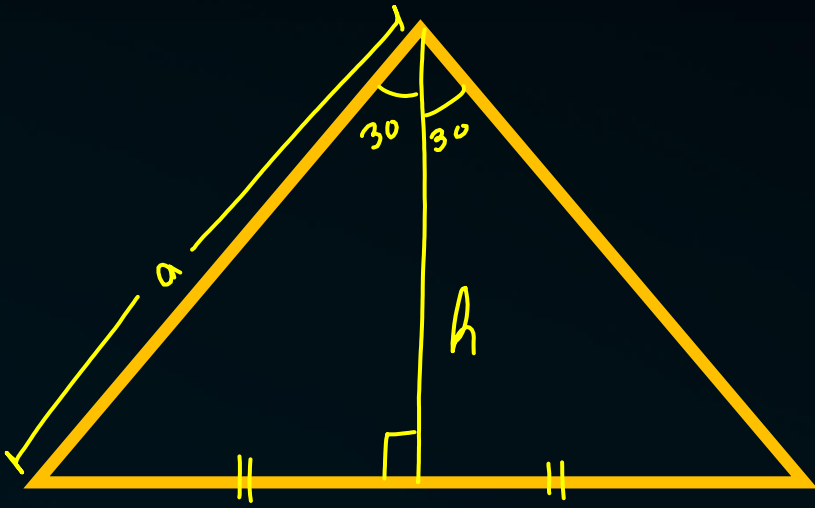
$$S = \frac{24}{2} = 12$$





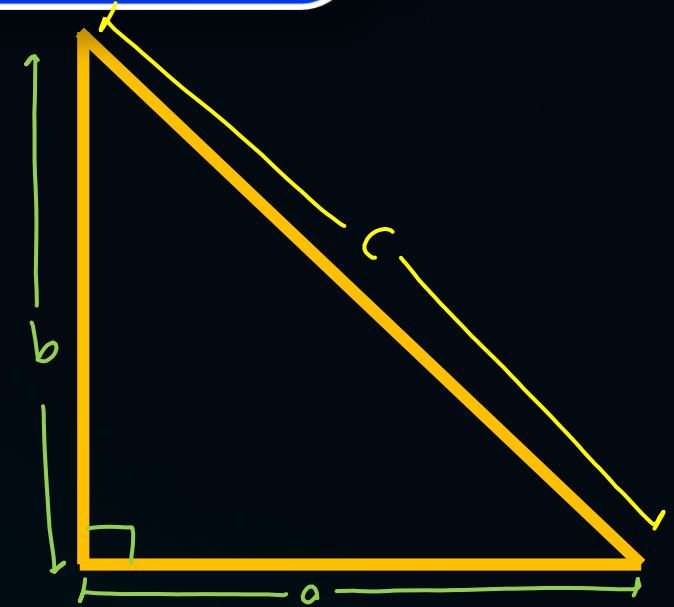
Topics : Right Triangle + Equilateral Triangle

Special Area formula for Equilateral and Right Triangle



$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

$$* \quad h = \frac{\sqrt{3}a}{2}$$



$$\text{Area} = \frac{1}{2} (a) (b)$$

$$a^2 + b^2 = c^2$$

Pythagoras Theorem



7, 24, 25	5	12	13	3, 4, 5
14, 48, 50	10	24	26	6, 8, 10
$7k, 24k, 25k$	15	36	39	9, 12, 15
	5k	12k	13k	15, 20, 25
$9k, 40k, 41k$				3k, 4k, 5k

Side And Height



$$\frac{\sqrt{3}a}{2} = h$$

$$\frac{1}{2}ah = \frac{\sqrt{3}}{4}a^2$$

$$h = \frac{\sqrt{3}a}{2}$$

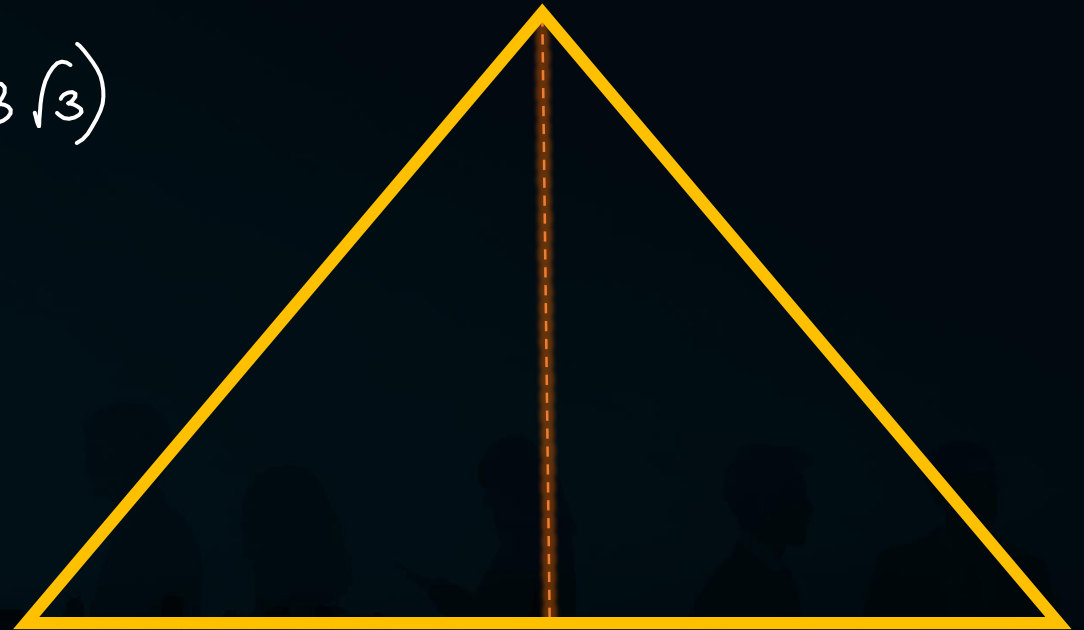
QUESTION - 12



#Q. Find the length of the perpendicular of Equilateral Triangle having side length as $3\sqrt{3}$ cm.

- A** 1.5 cm
- B** 4.5 cm ✓
- C** 6 cm
- D** $3\sqrt{3}$ cm

$$h = \frac{\sqrt{3}}{2} (3\sqrt{3})$$
$$= 4.5$$



QUESTION – 13 (CAT 2020)



$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

#Q. From the interior point of an equilateral triangle, perpendiculars are drawn on all three sides. The sum of the lengths of the perpendiculars is 'p'. Then the area of the triangle is

$$\ell + m + n = p$$

A $p^2/2\sqrt{3}$

B $2p^2/\sqrt{3}$

C $p^2/\sqrt{3}$ ✓✓

D $\sqrt{3}p^2/2$

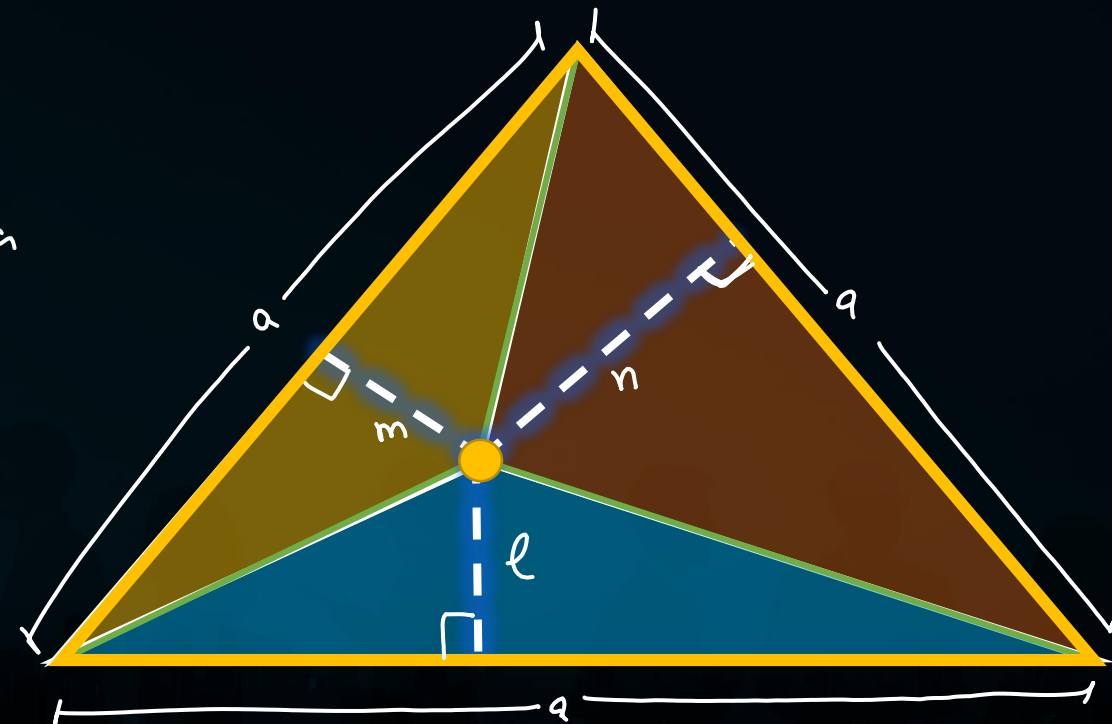
$$\frac{1}{2} a(m) + \frac{1}{2} a\ell + \frac{1}{2} an = \text{Area}$$

$$\frac{1}{2} a(\ell + m + n) = \text{Area}$$

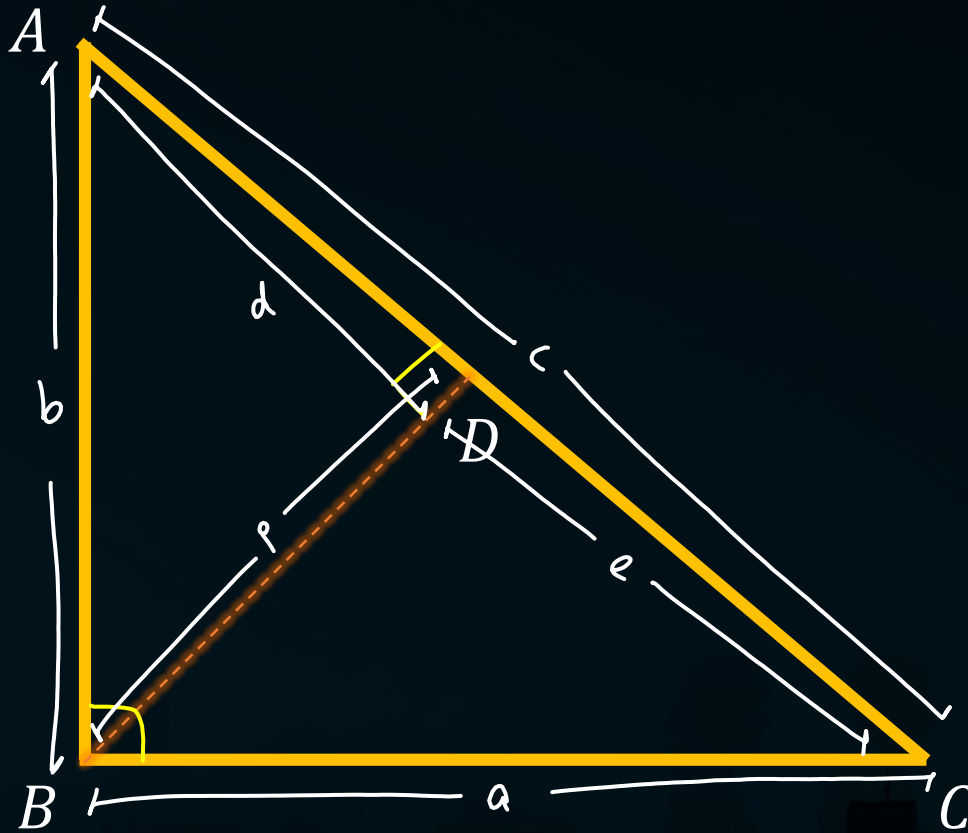
$$\frac{pa}{2} = \text{Area}$$

$$\frac{p^2 a^2}{4} = \text{Area}^2$$

$$p^2 \left[\frac{\text{Area}}{\sqrt{3}} \right] = \text{Area}$$



Important Figure



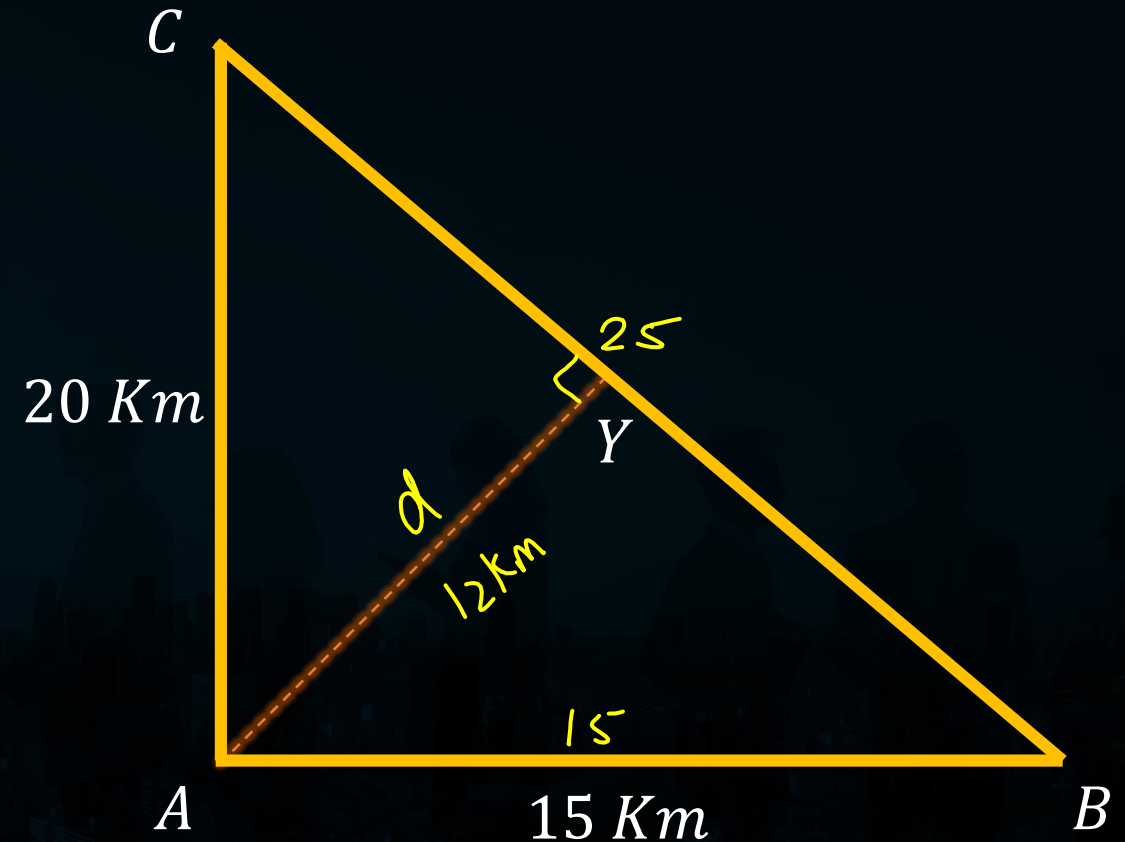
- ✓ $a^2 + b^2 = c^2$
- ✓ $pc = ab$
- ✓ $p^2 = de$

#Q. Let ABC be a right-angled triangle with BC as the hypotenuse. Lengths of AB and AC are 15 km and 20 km, respectively. The minimum possible time, in minutes, required to reach the hypotenuse from A at a speed of 30 km per hour is: (TITA)

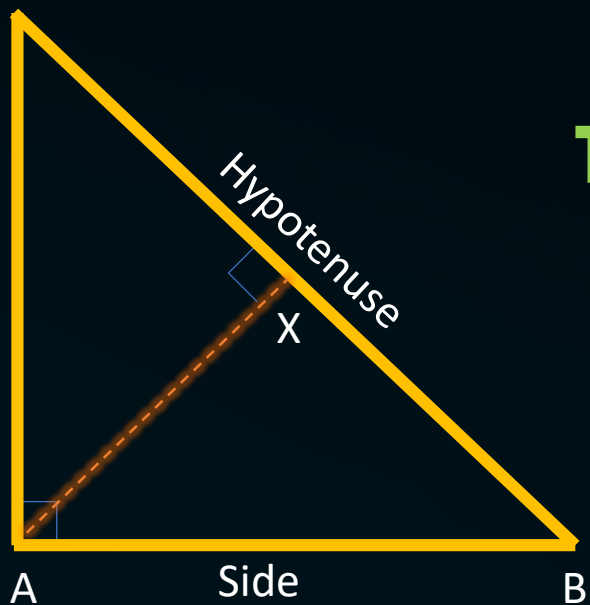
$$\begin{matrix} 3 & & 4 \\ (15) & (20) & = 25d \end{matrix}$$

$$d = 12$$

$$T = \frac{12}{30} \times 60 = 24 \text{ min}$$

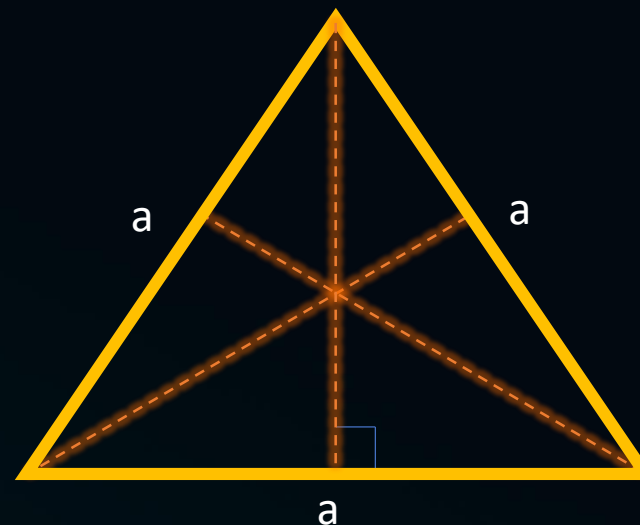


Right Triangle



- ❖ Area = $\frac{1}{2}$ (Product of Sides)
- ❖ $\frac{1}{2}$ (Product of Sides) = $\frac{1}{2}$ (hypotenuse)(AX)
- ❖ *Pythagoras Theorem*
 $3k, 4k, 5k$; $5k, 12k, 13k$; $7k, 24k, 25k$
 $9k, 40k, 41k$; $16k, 63k, 65k$

Equilateral Triangle

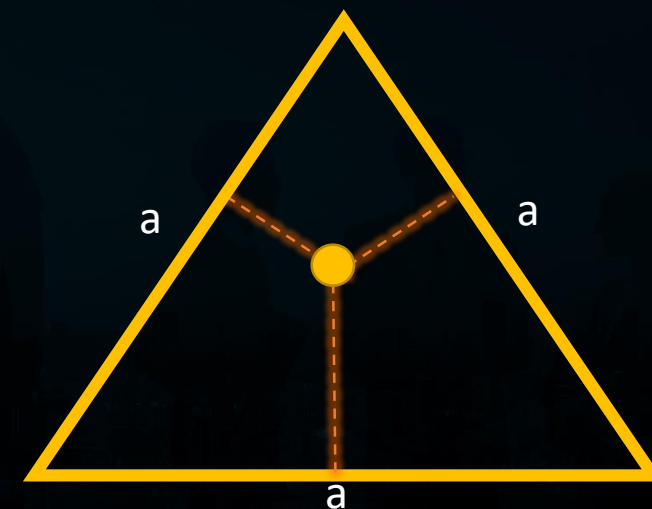


❖ Area = $\frac{\sqrt{3}}{4} a^2$

❖ $h = \frac{\sqrt{3}}{2} a$

❖ Area = $\frac{p^2}{\sqrt{3}}$

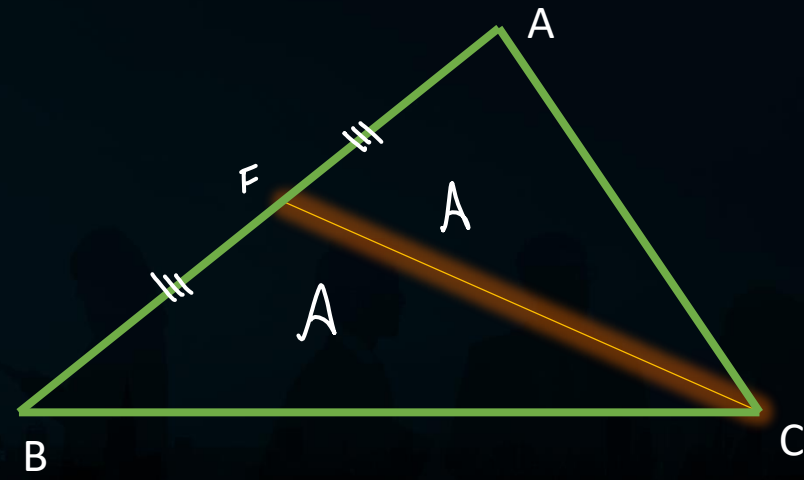
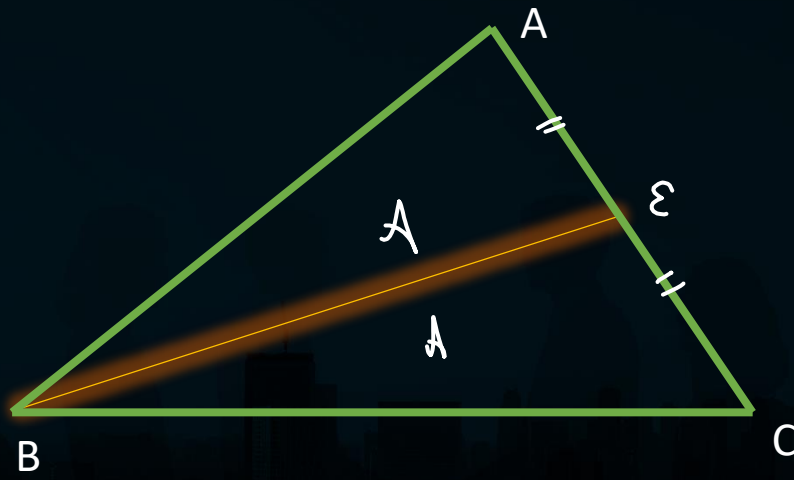
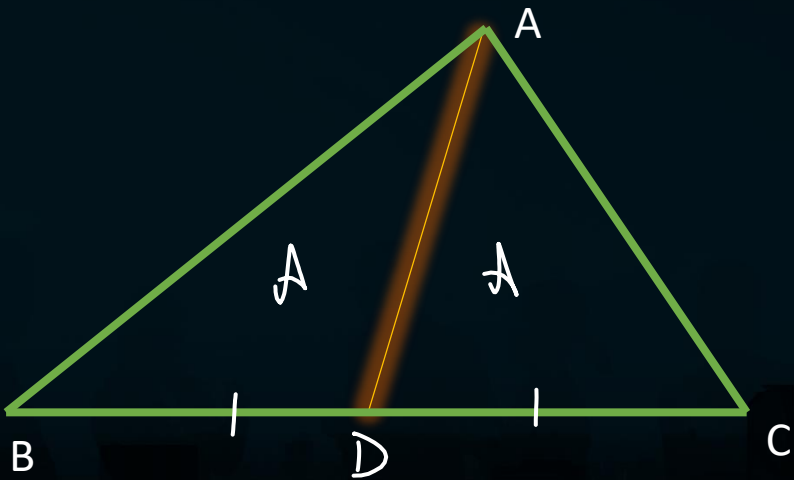
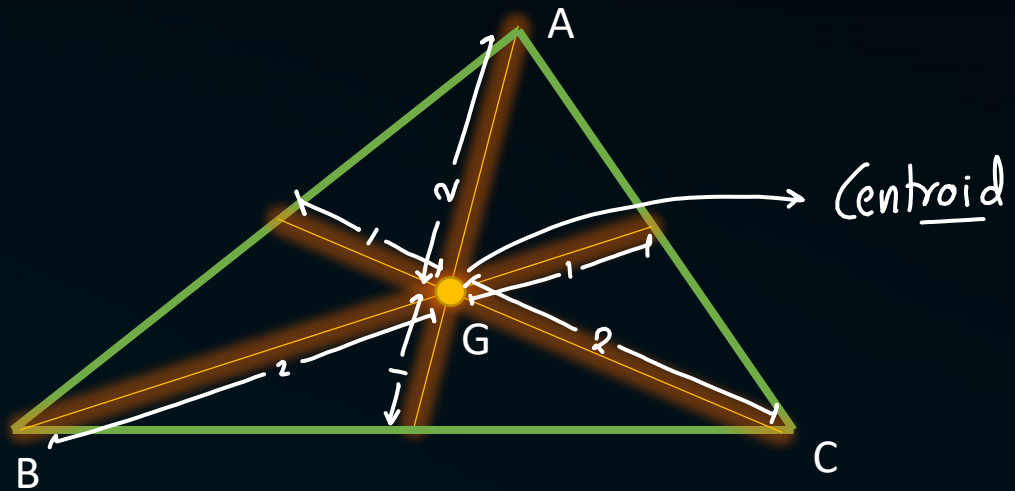
if $p = a + b + c$



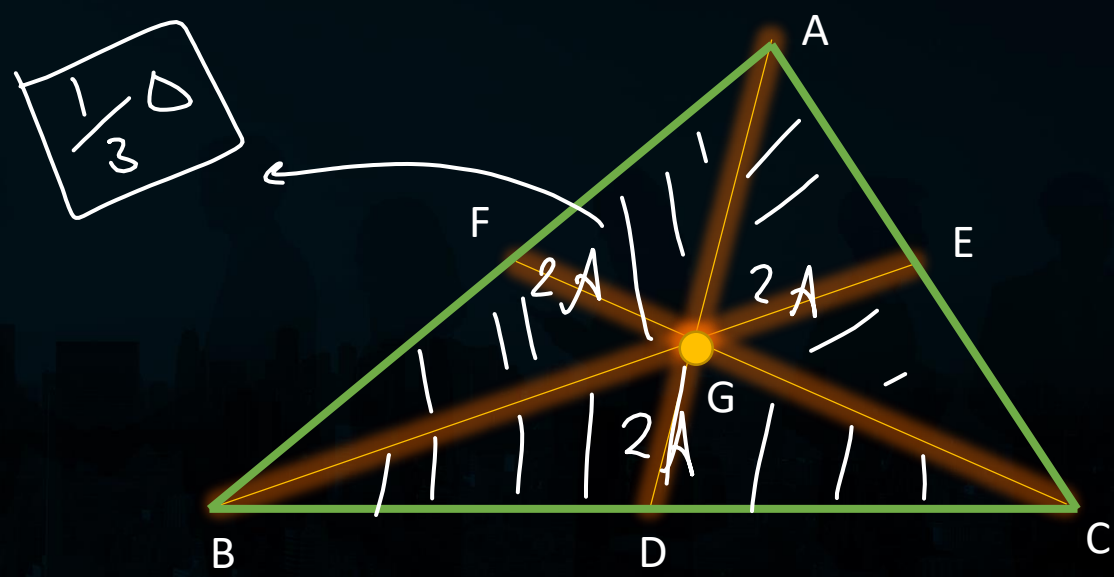
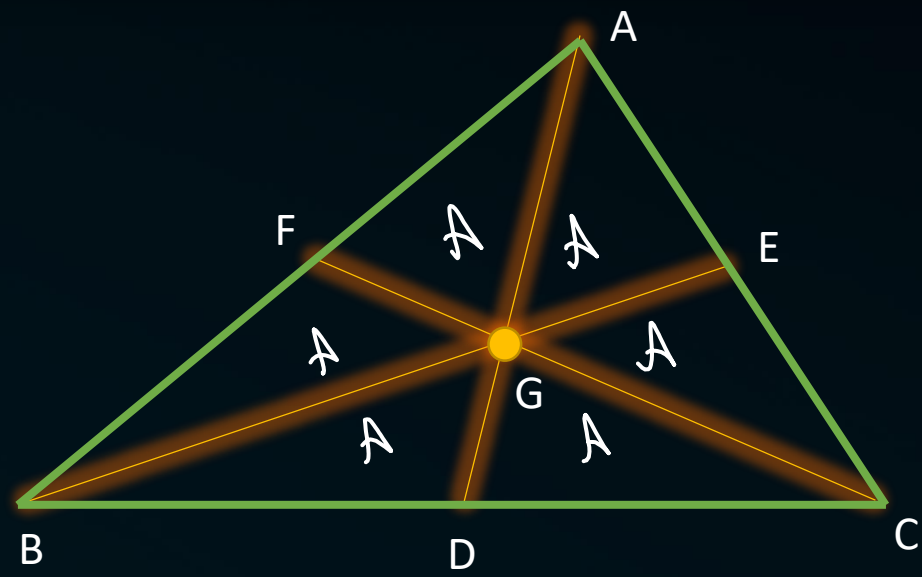


Topics : Important Lines and Centres

Medians and Centroid – Meaning & Important Ratios



Centroid – Division of Area



QUESTION - 15



#Q. In a Triangle ABC, AD is the median and area of ABD is 24 sq. units. Find the length of Perpendicular AM, if $BD = 6\text{cm}$.

A

6 cm

B

8 cm ✓✓

C

10 cm

D

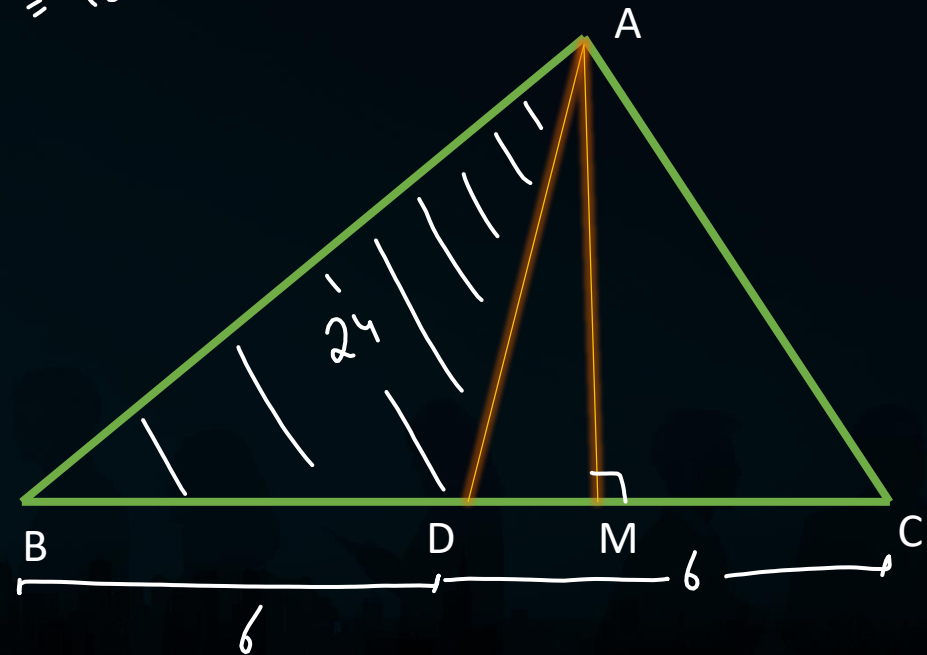
12 cm

Area of $\triangle ABC = 48$

$BC = 12$

$$\frac{1}{2}(\cancel{12})(AM) = \cancel{48}^8$$

$AM = 8$



QUESTION – 16 (CAT 2017)



#Q. From a triangle ABC with sides of lengths 40 ft, 25 ft and 35 ft, a triangular portion GBC is cut off where G is the centroid of ABC. The area, in sq ft, of the remaining portion of triangle ABC is:

- A** $225\sqrt{3}$
- B** $500/\sqrt{3}$ ✓✓
- C** $275/\sqrt{3}$
- D** $250/\sqrt{3}$

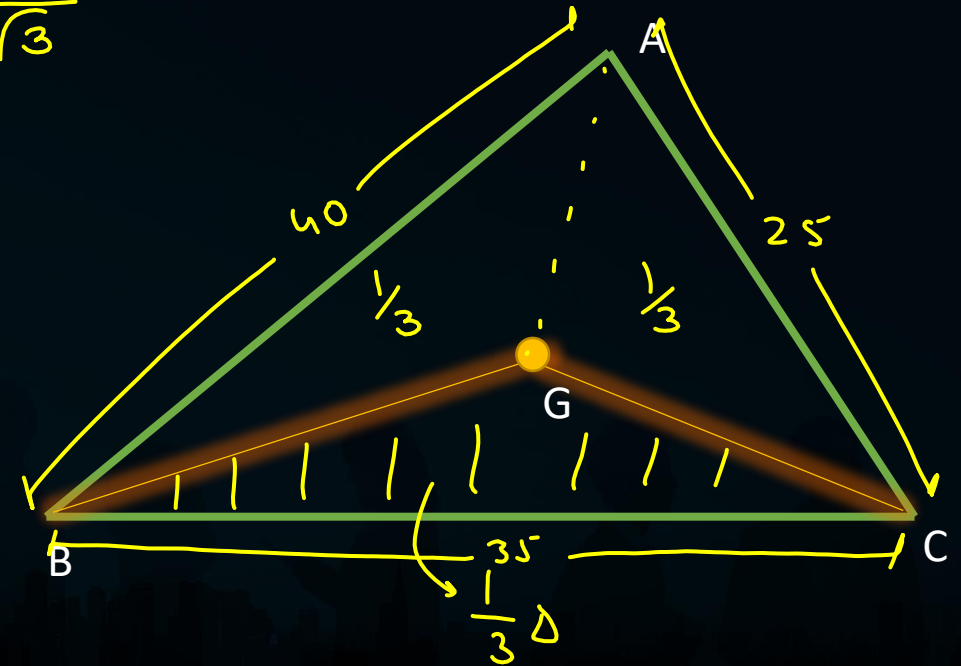
$$\frac{2}{3} \Delta = \frac{2}{3} \times 250\sqrt{3} = \frac{500}{\sqrt{3}}$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{40(10)(25)(\frac{15}{3})}$$

$$= 10 \times 5 \times 5 \sqrt{3}$$

$$= 250\sqrt{3}$$

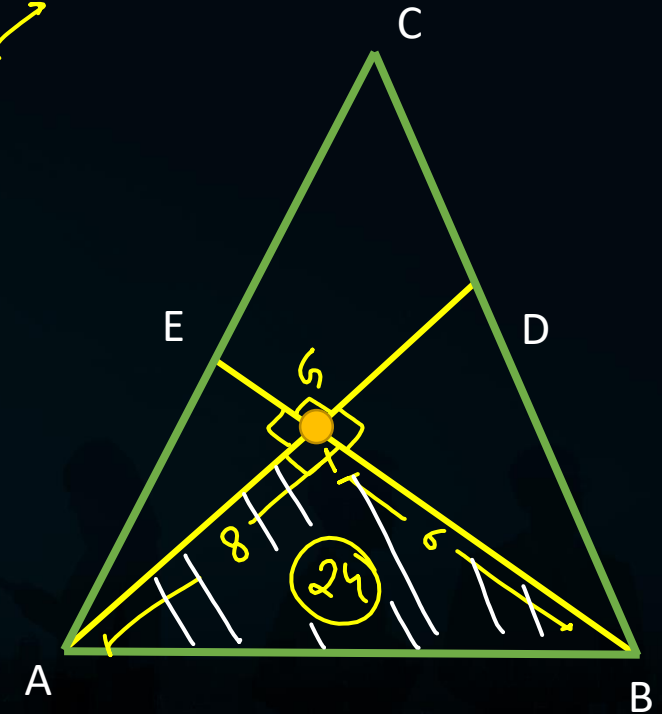
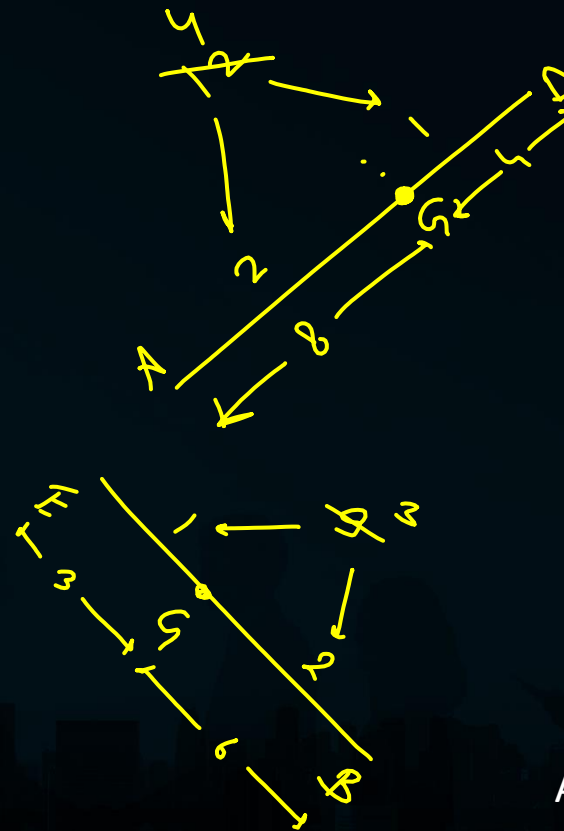


QUESTION – 17 (CAT 2019 – Slot 2)

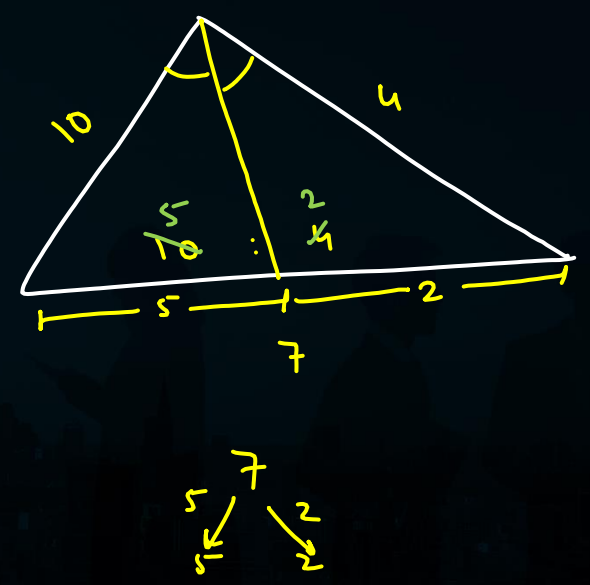
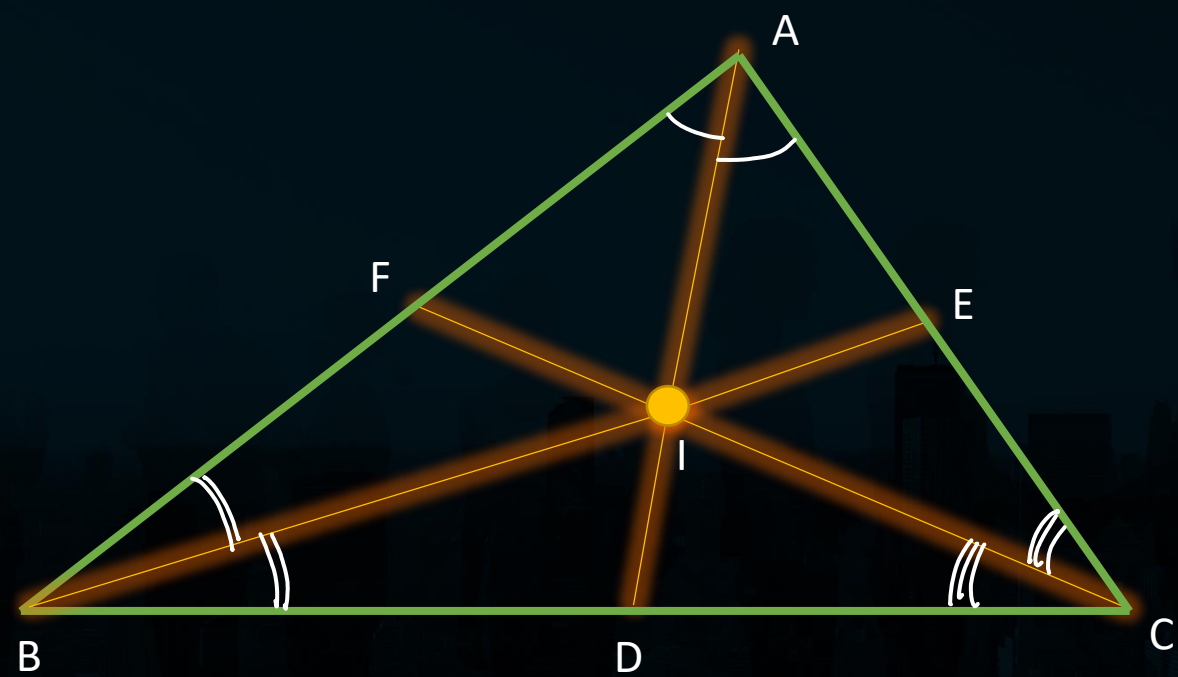
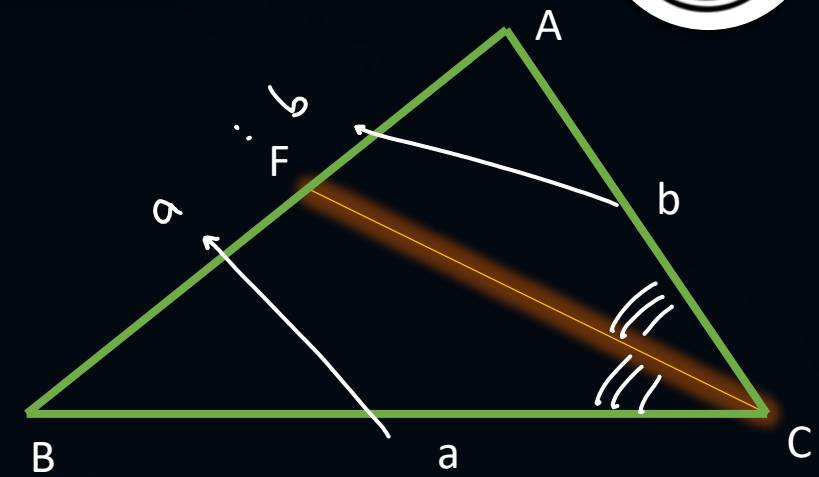
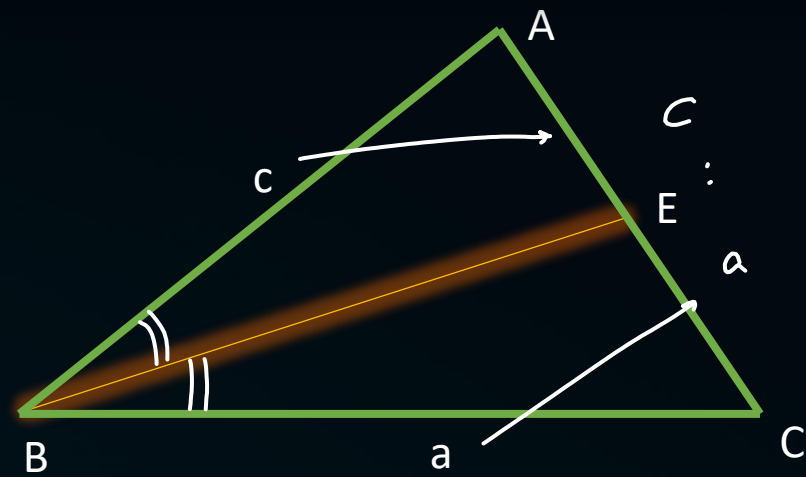
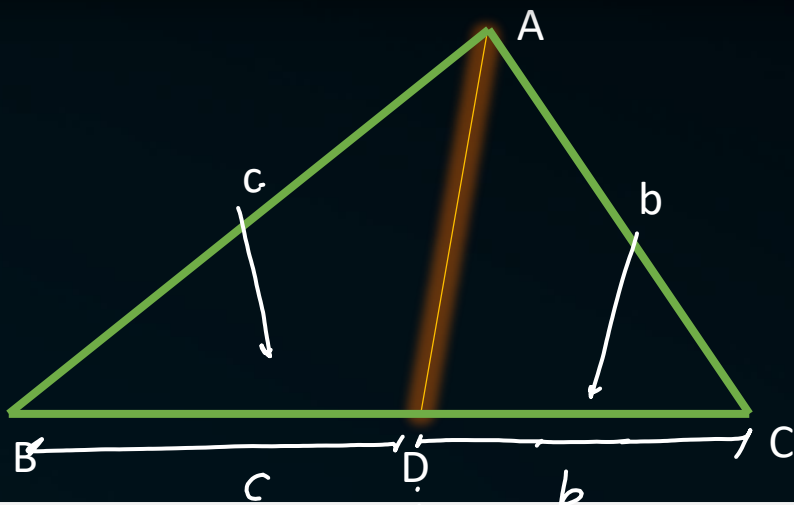


#Q. In a triangle ABC, medians AD and BE are perpendicular to each other, and have lengths 12 cm and 9 cm, respectively. Then, the area of triangle ABC, in sq cm, is

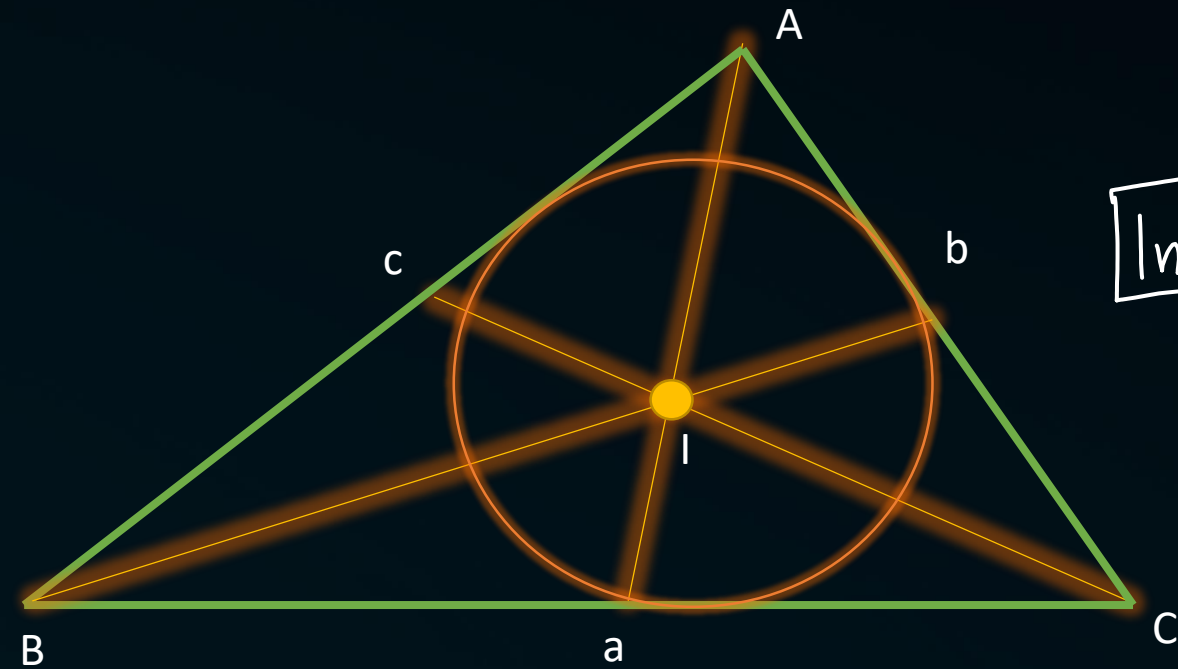
- A** 80 cm^2
- B** 68 cm^2
- C** 72 cm^2 ✓✓
- D** 144 cm^2



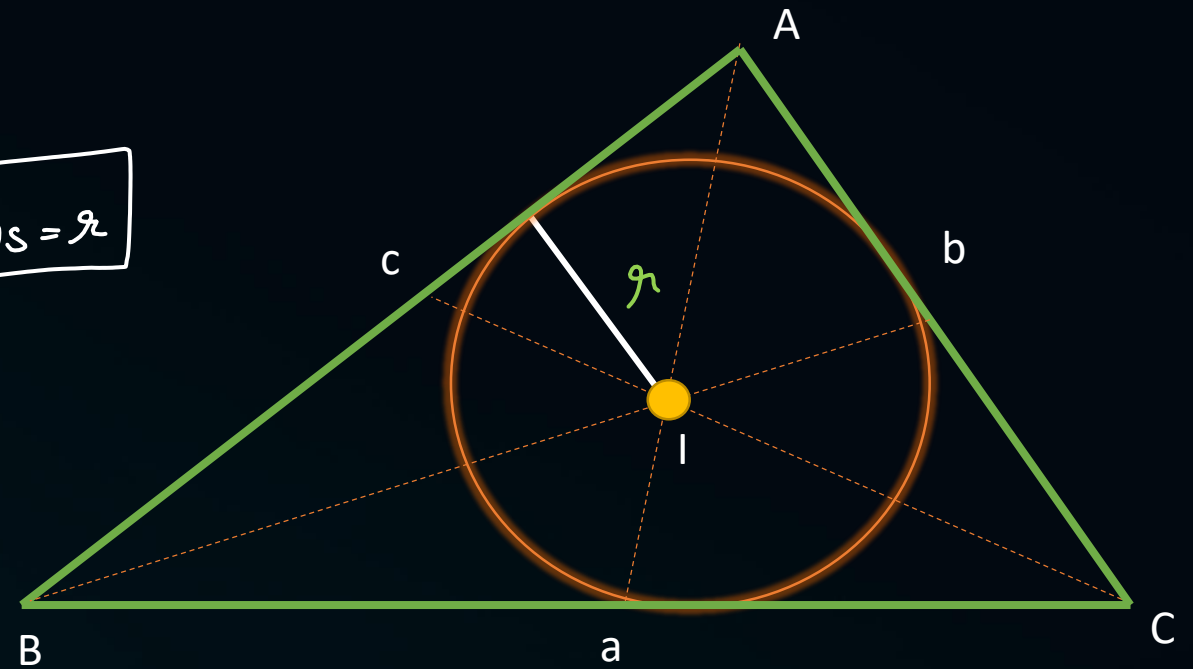
Angle Bisector and In centre Meaning



Area of Triangle Vs In radius



Inradius = r



$\Delta = rs$

Area of Δ

inradius

$r = \frac{\Delta}{s}$

Area of Δ

semi-peri:

r is same for all Δ

QUESTION – 18



#Q. If the inradius of the Triangle having the sides as 6 cm , 12 cm and 12cm is r. Find the value of $\sqrt{15} r$.

A

5

B

6

C

9 ✓

D

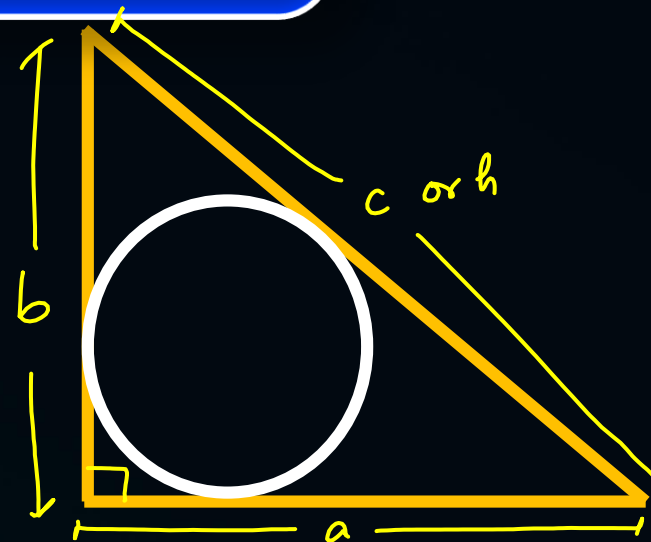
12

$$r = \frac{\Delta}{s} = \frac{\sqrt{15(9)(3)(3)}}{15} = \frac{9\sqrt{15}}{15}$$

$$\sqrt{15} r = \sqrt{15} \left[\frac{9\sqrt{15}}{15} \right]$$

$$= 9 \checkmark$$

2 Special Triangle & In radius



✓

$$r = \frac{a}{2\sqrt{3}}$$

✓

$$r = \frac{\Delta}{s} = \frac{\frac{\sqrt{3}}{4}a^2}{3 \times \frac{a}{2}}$$

✓

$$r = s - h$$

$$r = \frac{a+b-c}{2}$$

✓

$$\frac{a+b+c}{2} - c$$

QUESTION – 19



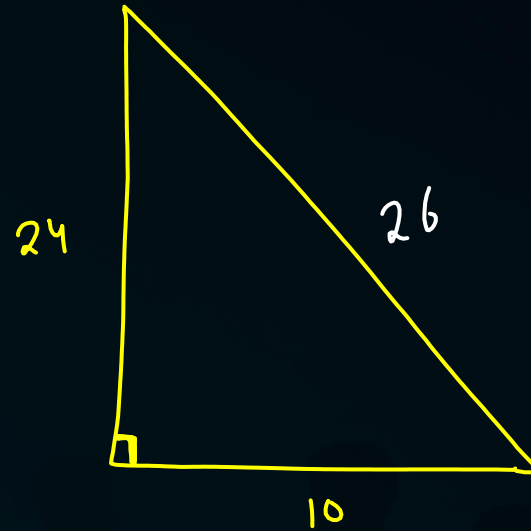
#Q. Find the inradius of the Right Triangle having the shorter sides as 10 cm and 24cm.

A 5 cm

B 6 cm

C 3 cm

D 4 cm ✓✓



$$r = \frac{\Delta}{s}$$

$$r = s - h$$

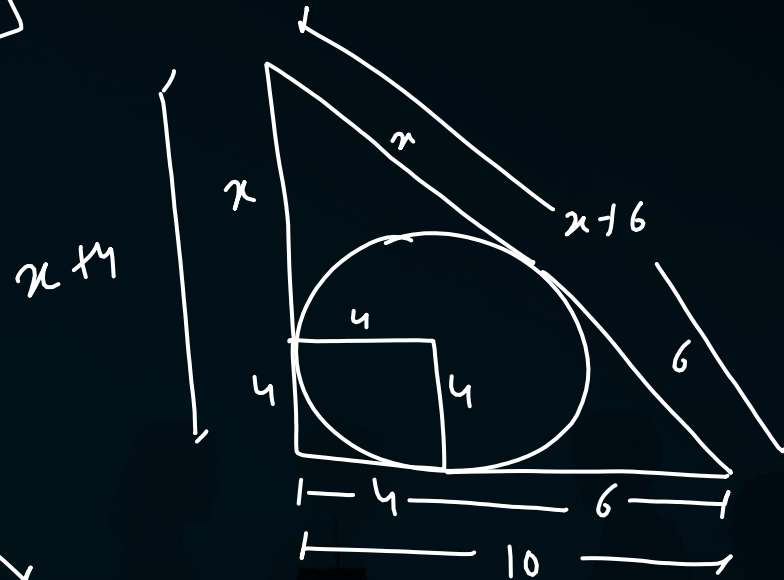
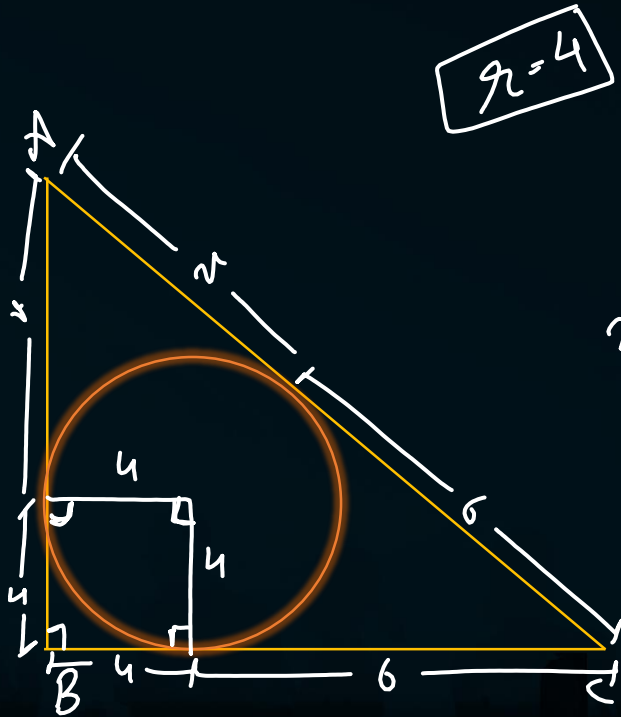
$$r = \frac{a+b-c}{2}$$

2024 Coordinate Geo

QUESTION – 20 (CAT 2021)



#Q. A circle of diameter 8 inches is inscribed in a triangle ABC where $\angle ABC = 90^\circ$. If $BC = 10$ inches then the area of the triangle in square inches is



$$(x+6)^2 = 10^2 + (x+4)^2$$

$$12x + 36 = 100 + 8x + 16$$

$$4x = 80$$

$$x = 20$$

$$\frac{1}{2} (10) (24) = 120 \checkmark$$

QUESTION – 21



#Q. A circle of diameter 8 inches is inscribed in a triangle ABC where $\angle ABC = 90^\circ$. If AC = 10 inches then the area of the triangle in square inches is



$$r = \frac{\Delta}{s}$$

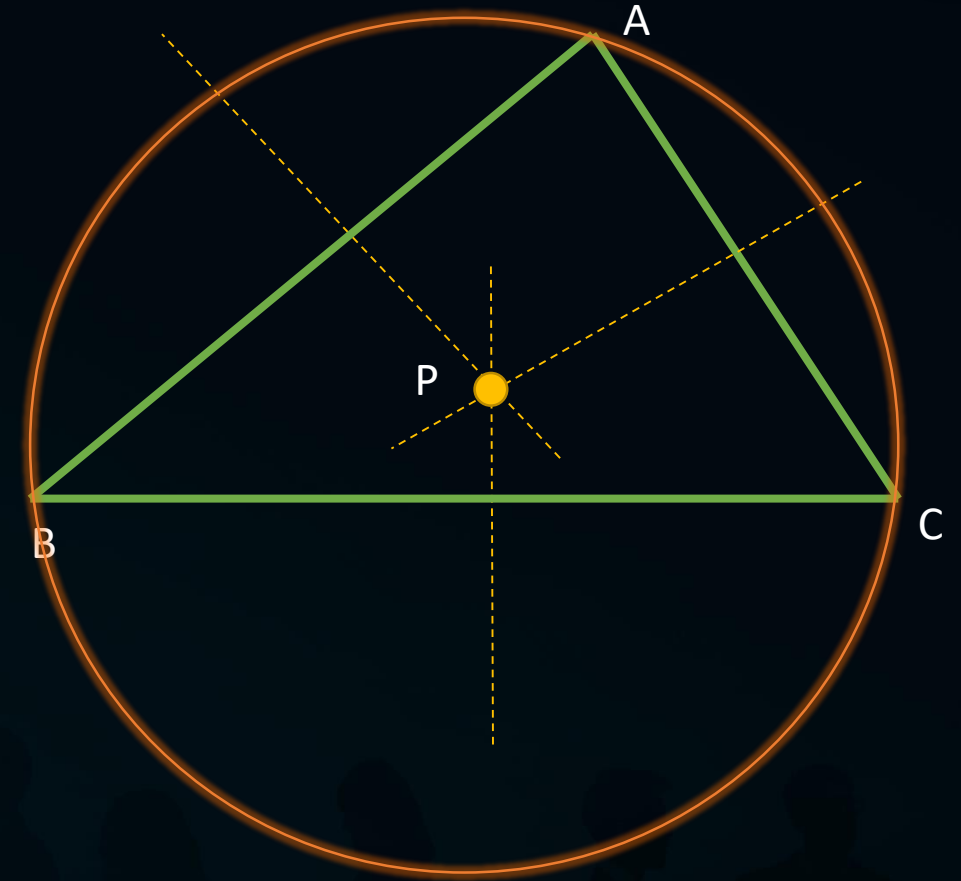
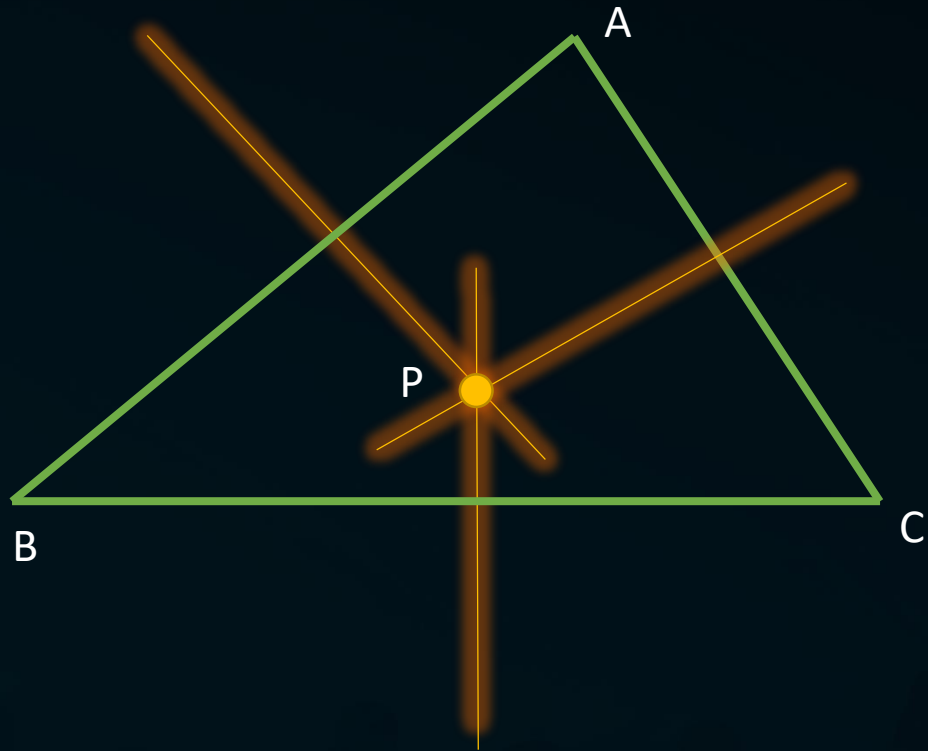
$$\checkmark r = 4$$
$$h = 10$$

$$r = s - h$$

$$\checkmark s = r + h = 4 + 10 = 14$$

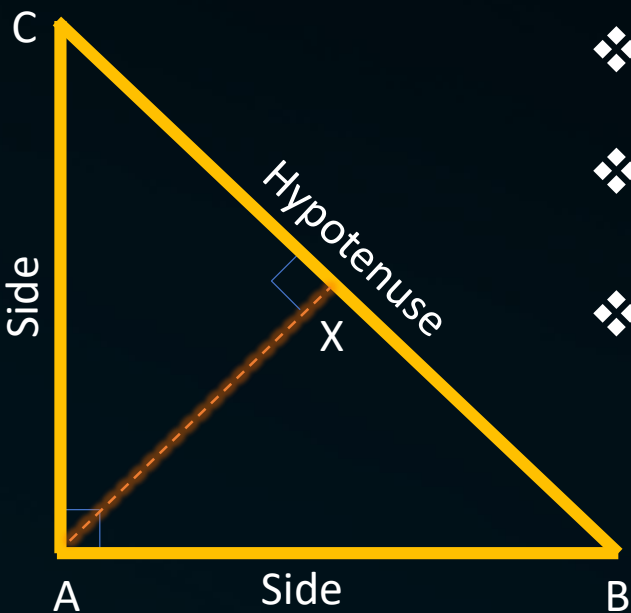
$$\begin{aligned} \text{Area} &= rs \\ &= 4 \times 14 \\ &= 56 \checkmark \end{aligned}$$

Perpendicular Bisectors and Circumcentre





Right Triangle



$$\diamond r = s - \text{hypotenuse}$$

$$\diamond r = \frac{a+b-c}{2}$$

$$\diamond R = \frac{\text{Hypotenuse}}{2}$$

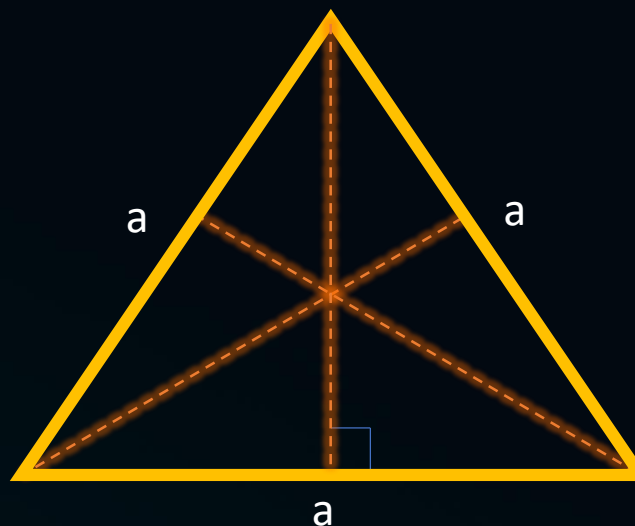
$$\diamond \text{Area} = \frac{1}{2} (\text{Product of Sides})$$

$$\diamond \frac{1}{2} (\text{Product of Sides}) = \frac{1}{2} (\text{hypotenuse})(AX)$$

$$\diamond \text{Pythagoras Theorem}$$

$$3k, 4k, 5k \quad ; \quad 5k, 12k, 13k \quad ; \quad 7k, 24k, 25k \\ 9k, 40k, 41k \quad ; \quad 16k, 63k, 65k$$

Equilateral Triangle



$$\diamond r = \frac{a}{2\sqrt{3}}$$

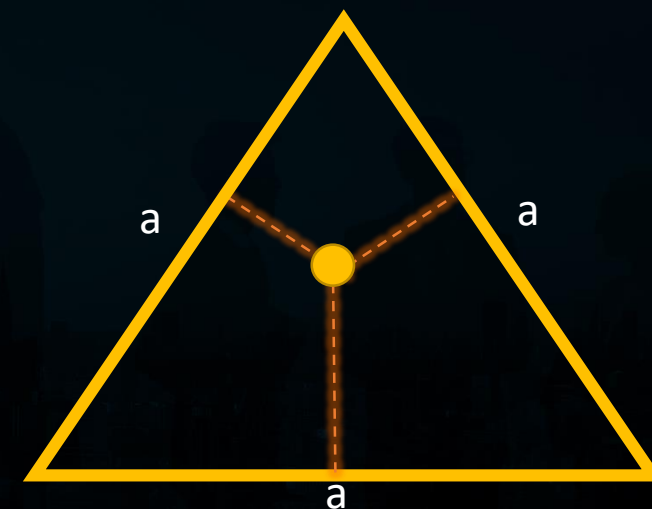
$$\diamond R = \frac{a}{\sqrt{3}}$$

$$\diamond \text{Area} = \frac{\sqrt{3}}{4} a^2$$

$$\diamond h = \frac{\sqrt{3}}{2} a$$

$$\diamond \text{Area} = \frac{p^2}{\sqrt{3}}$$

$$\text{if } p = a + b + c$$



SUMMARY



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ADVANCE TRIANGLES



THANK
You

